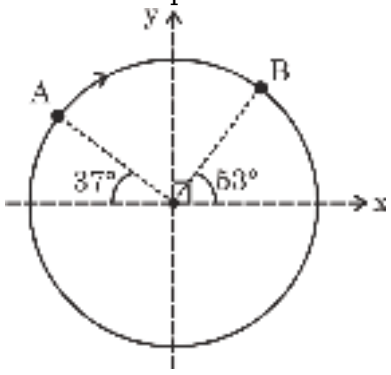
PHYSICS

1) A particle starts from A with constant speed v on the circle of radius R. Find the magnitude of average velocity from A to B:



(1) $\frac{2v}{\pi}$

(2) $\frac{2\sqrt{2}v}{3\pi}$

(3) $\frac{2\sqrt{2}v}{\pi}$

(4) $\frac{v}{\pi R}$

2) Two boys are standing at the ends A & B of a ground where AB = a. The boy at B starts running in a direction perpendicular to AB with velocity V_1. The boy at A starts running simultaneously with velocity V and catches the other boy in a time t, where t is :

(1) $\frac{a}{\sqrt{V^2 + V_1^2}}$

(2) $\sqrt{\frac{a^2}{(V^2 - V_1^2)}}$

(3) $\frac{a}{V - V_1}$

(4) $\frac{a}{V + V_1}$

3) A particle starts its motion from rest and traverses a distance □ with uniform acceleration, then moves uniformly over a further distance 2□ and finally comes to rest after moving a further distance 3□ under uniform retardation. Assuming entire motion to be rectilinear motion. The ratio of average speed over the journey to the maximum speed on its ways is:-

(1) $\frac{1}{5}$

(2) $\frac{2}{5}$

(3) $\frac{3}{5}$

(4) $\frac{4}{5}$

4) A projectile A is projected from ground. An observer B running on ground with uniform velocity of magnitude 'v' observes A to move along a straight line. The time of flight of A as measured by B is T. Then the range R of projectile on ground is :-

(1) R = vT

(2) R < vT

(3) R > vT

(4) information insufficient

5) Two trains one travelling at 15 ms^{-1} and other at 20 ms^{-1} are headed towards one another along a straight track. Both the drivers apply brakes simultaneously when they are 500 m apart. If each train has a retardation of 1 ms^{-2}, the separation between both trains after they stop is :-

(1) 192.5 m

(2) 225.5 m

(3) 187.5 m

(4) 155.5 m

6) A flywheel can rotate in order to store kinetic energy. The flywheel is a uniform disk made of a material with a density ρ and tensile strength σ (measured in Pascal), a radius r, and a thickness h, the flywheel is rotating at the maximum possible angular velocity so that it does not break. The maximum kinetic energy per kilogram that can be stored in the flywheel is $\propto \rho^x \sigma^y r^z$. Here α is a dimensionless constant find the value of x + y + z :

(1) 2

(2) 3

(3) 5

(4) Zero

7) Diameter of a steel ball is measured using a vernier callipers which has divisions of 0.1 cm on its main scale (MS) and 10 divisions of its vernier scale (VS) match 9 divisions on the main scale. Three such measurements for a ball are given as:

S.No.	MS(cm)	VS divisions
1	0.5	8
2	0.5	4
3	0.5	6

If the zero error is –0.03 cm, then mean corrected diameter is :

(1) 0.53 cm

(2) 0.56 cm

(3) 0.59 cm

(4) 0.52 cm

8) Determine the maximum acceleration in m/s^2 of the train in which a box lying on its floor will remain stationary, given that the co-efficient of static friction between the box and the train's floor is 0.15 :

(1) 3
(2) 1
(3) 1.5
(4) 2.5

9) A rope of length 5m is kept on frictionless surface and a force of 5 N is applied to one of its end. Find tension in the rope at 1 m from this end ..... N

(1) 1
(2) 3
(3) 4
(4) 5

10) A sphere of mass m moving with some velocity enters in a hanging bag of sand of mass M and stops. If the system (bag + sphere) raised by height h, then the velocity of the sphere was :

(1) $\frac{M+m}{m}\sqrt{2gh}$
(2) $\frac{M}{m}\sqrt{2gh}$
(3) $\frac{m}{M+m}\sqrt{2gh}$
(4) $\frac{m}{M}\sqrt{2gh}$

11) Match the column-I to column-II :

Column-I (body)		**Column-II (location of COM)**	
(P)	Uniform rod of length L	(1)	$\frac{3R}{8}$ from centre (on axis of symmetry)
(Q)	Uniform solid cone of height h	(2)	At L/2 from one end.
(R)	Uniform solid hemi sphere of radius R	(3)	h/4 from base (axis of symmetry)
(S)	Uniform semi-circular ring of radius R	(4)	$\frac{2R}{\pi}$ from centre (on axis of symmetry)

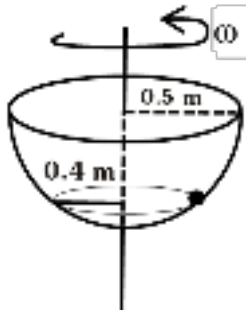
(1) P→2 ; Q→3 ; R→1 ; S→4

(2) P→1 ; Q→3 ; R→4 ; S→2

(3) P→2 ; Q→1 ; R→3 ; S→4

(4) P→2 ; Q→3 ; R→4 ; S→1

12) A particle of mass m is held at rest on the surface of a hemispherical bowl with respect to the bowl which is rotating with angular velocity ω. Radius of bowl is 0.5 m. According to the figure radius of the circle of the particle is 0.4 m. Find angular speed ω of the bowl : (all surfaces are smooth)



(1) $\frac{10}{\sqrt{3}}$ rad/s

(2) $\sqrt{\frac{10}{3}}$ rad/s

(3) $\sqrt{20}$ rad/s

(4) $\sqrt{10}$ rad/s

13) A pendulum bob has a speed of $3ms^{-1}$ at its lowest position. The pendulum is 0.5 m long. The speed of the bob when the length makes an angle of 60° to the vertical.

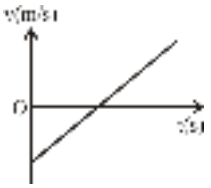
(1) $3ms^{-1}$

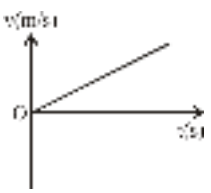
(2) $\frac{1}{3}ms^{-1}$

(3) $\frac{1}{2}ms^{-1}$

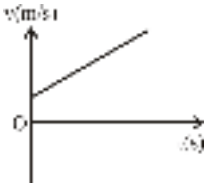
(4) $2ms^{-1}$

14) A particle moves along x-axis in such a way that its x-coordinate varies with time t according to the equation $x = (8 - 4t + 6t^2)$ metre.The velocity of the particle will vary with time according to the graph :-

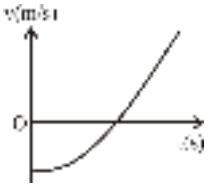
(1) 

(2) 

(3)



(4)



15) A body freely falling from the rest has a velocity 'u' after it falls through a height 'h'. The distance it has to fall down for its velocity to become double, is

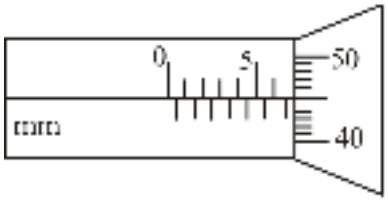
(1) 2h

(2) 4h

(3) 6h

(4) 8h

16) Read the screw gauge
* Main scale has 1/2 mm marks
* Circular scale has 50 divisions. * In complete rotation, the screw advanced by 1/2 mm



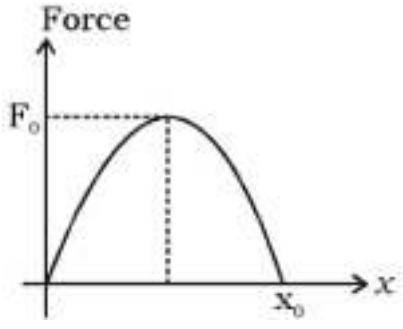
(1) 6.90 mm

(2) 6.95 mm

(3) 6.93 mm

(4) 6.97 mm

0



17) Work done in displacement x will be (graph is semi ellipse)-

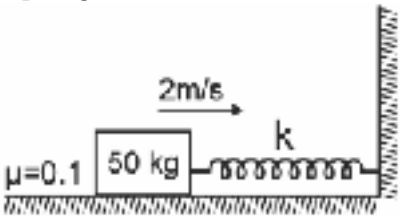
(1) $\frac{\pi x_0 F_0}{2}$

(2) $\frac{\pi x_0 F_0}{3}$

(3) $\frac{\pi x_0 F_0}{4}$

(4) $\pi x_0 F_0$

18) A block of mass 50 kg is projected horizontally on a rough horizontal floor. The coefficient of friction between the block and the floor is 0.1. The block strikes a light spring of stiffness k = 100 N/m with a velocity 2m/s. The maximum compression of the spring is :



(1) 1 m

(2) 2 m

(3) 3 m

(4) 4 m

19) A pump is used to deliver water through a pipe at constant rate. To obtain water at four times the previous rate area of cross-section of pipe is doubled. Find by what amount the power should be increased. :-

(1) 2 times

(2) 4 time

(3) 8 times

(4) 16 times

20) Find force at (–1, 1, –2) if potential energy for a particle is, $U = x^2 + 2y^2 + z^2$?

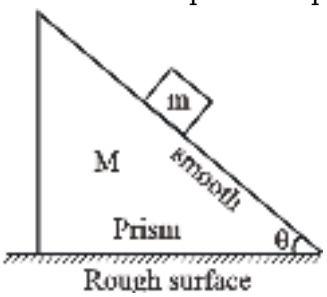
(1) $2\hat{i} - 4\hat{j} + 4\hat{k}$

(2) $-\hat{i} - 2\hat{j} - \hat{k}$

(3) $\hat{i} + \hat{j} + 3\hat{k}$

(4) $5\hat{i} - \hat{j} + 3\hat{k}$

21) A body of mass m is coming down the inclined prism. If prism doesn't move, the frictional force between prism and ground will be :-



(1) $\frac{1}{2}$ mg sin 2θ

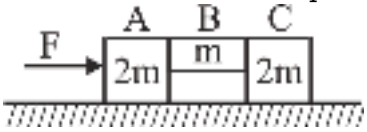
(2) $\frac{(M+m)g\sin\theta}{2}$

(3) $\frac{(M+m)g\cos\theta}{2}$

(4) Mg sin θ

22) The system is pushed by a force F as shown in figure. All surfaces are smooth except between B

and C. Friction coefficient between B and C is μ. Minimum value of F to prevent block B from downward slipping is :-



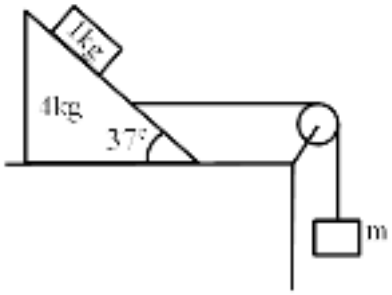
(1) $\left(\frac{3}{2\mu}\right)mg$

(2) $\left(\frac{5}{2\mu}\right)mg$

(3) $\left(\frac{5}{2}\right)\mu mg$

(4) $\left(\frac{3}{2}\right)\mu mg$

23) If all surfaces are frictionless then what will be the value of mass 'm' so that 1 kg block does not slip:



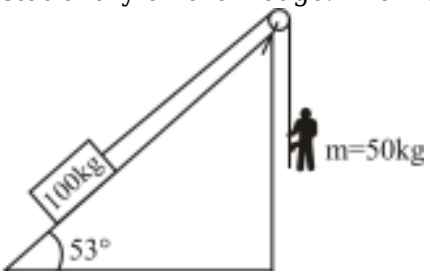
(1) 5 kg

(2) 10 kg

(3) 15 kg

(4) None

24) In the given figure by what acceleration the boy must go up so that 100 kg block remains stationary on the wedge. The wedge is fixed and friction is absent everywhere (g = 10 m/s^2) :-



(1) 2 m/s^2

(2) 4 m/s^2

(3) 6 m/s^2

(4) 8 m/s^2

25)

A ball is dropped from a height h on a floor of coefficient of restitution e. The total distance covered by ball before second hit is :-

(1) $h(1 - 2e^2)$

(2) $h(1 + 2\ e^2)$

(3) $h(1 + e^2)$

(4) he^2

26) Point masses 1,2,3 and 4 kg are lying at the point (0, 0, 0), (2, 0, 0), (0, 3, 0) and (–2, –2, 0) respectively. The moment of inertia of this system about x-axis will be :-

(1) 43 kg-m^2

(2) 34 kg-m^2

(3) 27 kg-m^2

(4) 72 kg-m^2

27) If rotational kinetic energy of rotating body is decreased by 51%. Then percentage change in angular momentum is :-

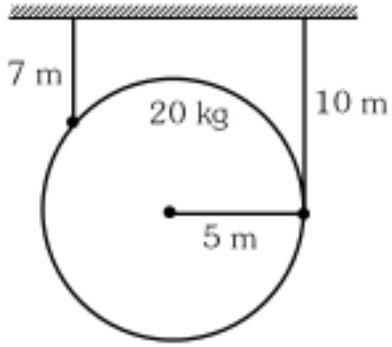
(1) 70%

(2) 30%

(3) 49%

(4) 51%

28) A disc of mass 20 kg is hanged using two strings of length 10 m and 7 m as shown in the figure.



The radius of the disc is 5m. Find tension in the string of length 7 m :-

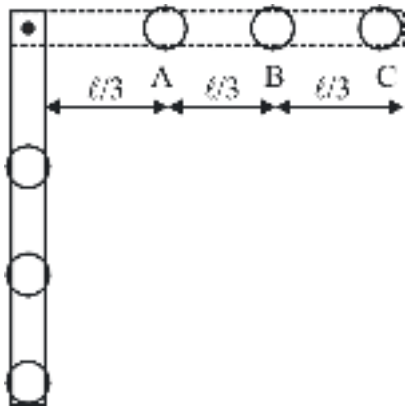
(1) $\frac{2000}{9}$N

(2) $\frac{1000}{6}$N

(3) $\frac{800}{9}$N

(4) $\frac{1000}{9}$N

29) A light rod carries three equal masses A, B and C as shown if fig. What will be velocity of B in vertical position of rod, if it released from horizontal. Position as shown in fig. ?



(1) $\sqrt{\frac{4g\ell}{7}}$

(2) $\sqrt{\frac{8g\ell}{7}}$

(3) $\sqrt{\frac{2g\ell}{7}}$

(4) $\sqrt{\frac{10g\ell}{7}}$

30) **Assertion (A) :** A couple does not exert a net force on an object even though it exerts a torque.
Reason (R) : Couple is a pair of two forces with equal magnitude but opposite directions acting simultaneously on a body in different lines of action.

(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**

(2) **(A)** is correct but **(R)** is not correct

(3) **(A)** is incorrect but **(R)** is correct

(4) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**

31) **Statement-I :** Kepler's second law is equivalent to conservation of linear momentum.
Statement-II : According to Kepler's second law, a line joining planet to the Sun sweeps out equal areas in equal intervals of time.

(1) Both statement-I and II are correct

(2) Both statement-I and II are incorrect

(3) Only statement-I is correct

(4) Only statement-II is correct

32) With what angular velocity the earth should spin in order that a body lying at 37° latitude may become weightless :-

(1) $\frac{5}{4}\sqrt{\frac{g}{R}}$

(2) $\frac{25}{16}\sqrt{\frac{g}{R}}$

(3) $\frac{5}{3}\sqrt{\frac{g}{R}}$

(4) $\frac{25}{9}\sqrt{\frac{g}{R}}$

33) Work done in launching a geo-stationary satellite (orbital radius 7R) of mass m from earth's surface (at rest) is :-

(1) $\frac{GMm}{14R}$
(2) $\frac{13}{14}\frac{GMm}{R}$
(3) $\frac{GMm}{7R}$
(4) $\frac{6}{7}\frac{GMm}{R}$

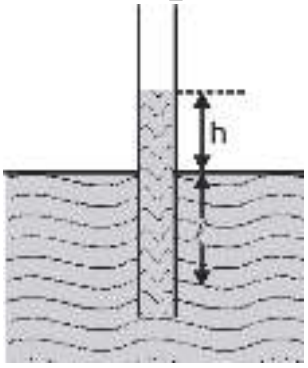
34) At which depth from earth surface the acceleration due to gravity become 40% of its value on the surface of earth :-

(1) 3200 km
(2) 2560 km
(3) 3840 km
(4) 1920 km

35) The escape velocity from the earth surface is 11.2 km/s, the mass of another planet is 100 times that of earth and its radius is 4 times that of the radius of earth. Escape velocity for the surface of this planet is :-

(1) 56.0 km/s
(2) 280 km/s
(3) 112 km/s
(4) 11.2 km/s

36) The capillary tube is dipped in water vertically. It is sufficiently long so that water rises to maximum height h in the tube. The length of the portion immered in water is □ > h. The lower end of



the tube is closed, the tube is taken out and opened again. Then-

(1) all water flow out of the tube.
(2) the length of the water column remaining in the tube will be h.
(3) the length of the water column remaining in the tube will be 2h
(4) the length of the water column remaining in the tube will be □ + h

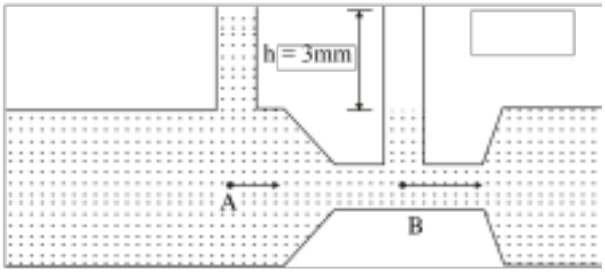
37) A cube of ice has a iron piece frozen inside. The cube floats in a beaker filled with water, when the ice melts then, the level of water in beaker. :-

(1) remains same

(2) level decreases

(3) level increases

(4) first increases then decreases

38) Cross sectional area at A and B are respectively 4 cm^2 and 2cm^2. If h = 3 mm then find rate of flow in given venturimeter 

(1) 5.6×10^{-5} m^3/s

(2) 130×10^{-6} m^3/s

(3) 170×10^{-6} m^3/s

(4) 162×10^{-6} m^3/s

39) There is a very small hole near the bottom of an open tank filled with a liquid. The speed of the water ejected does not depend on (one or more options may be correct)
A. Area of the hole
B. Density of the liquid
C. Height of the liquid from the hole
D. Acceleration due to gravity

(1) A, C

(2) A, B

(3) B, C

(4) B, D

40) A rubber rope is 4 m long and is suspended from ceiling. Find increment in its length due to its own weight.
(Young's modulus Y = 5×10^8 N/m^2)
(Density $\rho = 0.75 \times 10^3$ Kg/m^3)

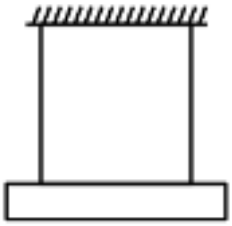
(1) 34 m

(2) 345 m

(3) 3.45×10^{-2} m

(4) 1.2×10^{-4} m

41) Two wires of equal length and cross-section area suspended as shown in figure. Their young's



modulus are Y_1 and Y_2 respectively. The equivalent young's modulus will be:

(1) $Y_1 + Y_2$

(2) $\frac{Y_1 + Y_2}{2}$

(3) $\frac{Y_1 Y_2}{Y_1 + Y_2}$

(4) $\sqrt{Y_1 Y_2}$

42) What is the velocity v of a metallic ball of radius r falling in a tank of liquid at the instant when its acceleration is one-half that of a freely falling body? (The density of metal and of liquid are ρ and σ respectively and the viscosity of the liquid is η) :-

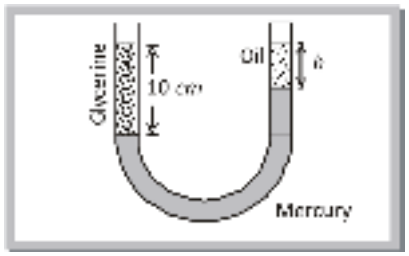
(1) $\frac{r^2 g}{9\eta}(\rho - 2\sigma)$

(2) $\frac{r^2 g}{9\eta}(2\rho - \sigma)$

(3) $\frac{r^2 g}{9\eta}(\rho - \sigma)$

(4) $\frac{2r^2 g}{9\eta}(\rho - \sigma)$

43) A vertical U-tube of uniform inner cross section contains mercury in both sides of its arms. A glycerin (density = 1.3 g/cm^3) column of length 10 cm is introduced into one of its arms. Oil of density 0.8 gm/cm^3 is poured into the other arm until the upper surfaces of the oil and glycerin are in the same horizontal level. Find the length of the oil column, Density of mercury = 13.6 g/cm^3



(1) 10.4 cm

(2) 8.2 cm

(3) 7.2 cm

(4) 9.6 cm

44) A particle is projected with velocity v_0 along x–axis. The deceleration on the particle is proportional to the square of the distance from the origin i.e., $a = -\alpha x^2$. The distance at which the particle stops is :-

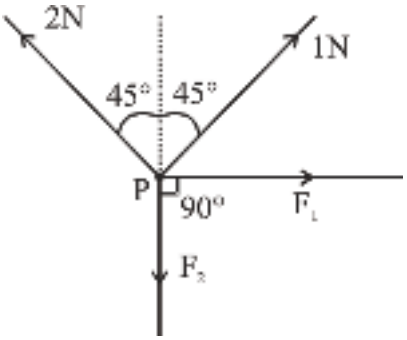
(1) $\sqrt{\frac{3v_0}{2\alpha}}$

(2) $\left(\frac{3v_0}{2\alpha}\right)^{\frac{1}{3}}$

(3) $\sqrt{\frac{3v_0^2}{2\alpha}}$

(4) $\left(\frac{3v_0^2}{2\alpha}\right)^{\frac{1}{3}}$

45) There are four forces acting at a point P produced by strings as shown in figure, which is at rest. The forces F_1 and F_2 are :-



(1) $\frac{1}{\sqrt{2}}N, \frac{3}{\sqrt{2}}N$

(2) $\frac{3}{\sqrt{2}}N, \frac{1}{\sqrt{2}}N$

(3) $\frac{1}{\sqrt{2}}N, \frac{1}{\sqrt{2}}N$

(4) $\frac{3}{\sqrt{2}}N, \frac{3}{\sqrt{2}}N$

CHEMISTRY

1) In which of the following promotion of e^- is not required for bonding :-

(1) C_2H_2

(2) SiO_2

(3) CO

(4) SF_6

2) Find the number of species in which atleast 2 types of chemical bond is present ?
KI, KHF_2, $CuSO_4.5H_2O$, NH_4Cl, Na_2O, Na_2O_2, CaC_2

(1) 3

(2) 4

(3) 5

(4) 7

3) Which of the following reaction is not possible ?

(1) $(CH_3)_2O + BF_3 \rightarrow [(CH_3)_2O \rightarrow BF_3]$

(2) $NH_3 + BF_3 \rightarrow [NH_3 \rightarrow BF_3]$

(3) $(CH_3)_2O + BF_4^- \rightarrow [(CH_3)_2O \rightarrow BF_4^-]$

(4) $N(CH_3)_3 + B(CH_3)_3 \rightarrow [N(CH_3)_3 \rightarrow B(CH_3)_3]$

4) In the process $N_2 \rightarrow N_2^+$ and $O_2 \rightarrow O_2^+$

(1) Both N-N and O-O bond get weakens

(2) N_2 becomes diamagnetic

(3) bond order of O_2 decreases

(4) In $O_2 \rightarrow O_2^+$ paramagnetism decreases

5) In which of the following species pπ-dπ bond is present but pπ-pπ bond is absent ?

(1) SO_3

(2) CS_2

(3) SO_2

(4) SO_2Cl_2

6) Which of the following sets of species is isostructural, planar and non polar :-

(1) ClF_3, BrF_3, XeO_3

(2) ClO_4^-, SO_4^{2-}, PO_4^{3-}

(3) $I_3^{\oplus}$, $XeF_3^{\oplus}$, S_3^{2-}

(4) I_3^-, $HgCl_2$, XeF_2

7) Which is correct reason for non-existence

Non-existing molecule	**Reason**
(1) XeF_5	odd e^- species
(2) PbI_4	Pb^{+4} is reducing agent
(3) PH_5	d-orbital is not present
(4) F_3^-	Steric hinderance

(1) (1)

(2) (2)

(3) (3)

(4) (4)

8)

In XeF_6, the number of lone pair on central atom, bond pair are respectively & it's hybridisation is-

(1) 0, 6, sp^3d^2

(2) 1, 6, sp^3d^3

(3) 0, 5, sp^3d

(4) 1, 5, sp^3d^2

9) The electronegativity of the following elements increases in the order :

(1) C < N < Si < P

(2) Si < P < C < N

(3) N < C < P < Si

(4) C < Si < N < P

10) Which of the following is an amphoteric hydroxide ?

(1) $Sr(OH)_2$

(2) $Ca(OH)_2$

(3) $Mg(OH)_2$

(4) $Be(OH)_2$

11) Electron gain enthalpy will be positive in

(1) O^{-2} is formed from O^{-1}

(2) O^{-1} is formed from O

(3) S^{-1} is formed from S

(4) Na^- is formed from Na

12) **Statement-1 :** order of ionisation energy F > Cl > Cl^- > F^-
Statement-2 : Size of Cl^- is smaller than F^-

(1) Statement (1) & (2) both are correct.

(2) Statement (1) & (2) both are incorrect.

(3) Statement (1) is correct but statement (2) is incorrect

(4) Statement (1) is incorrect but statement (2) is correct

13) If first four IP values of an element are 15 , 190 , 210 , 230 ev / atom then element is

(1) Al

(2) Si

(3) Na

(4) Mg

14) Which of the following process is highest exothermic

(1) $P_{(g)} \rightarrow P_{(g)}$

(2) $O_{(g)} \rightarrow O^-_{(g)}$

(3) $Si_{(g)} \rightarrow Si^{-}_{(g)}$

(4) $S_{(g)} \rightarrow S^{-}_{(g)}$

15) Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R),
Assertion (A) : Cl^{-} and Ca^{+2} both are isoelectronic species.
Reason (R) : Isoelectronic species must have same atomic number.

(1) Both (A) and (R) are true and (R) is not the correct explanation of (A)

(2) Both (A) and (R) are true and (R) is the correct explanation of (A)

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

16) Which of the following is correct electronic configuration for s-block.

(1) [Kr] $4d^{10}$ $5s^2$

(2) [Xe] $4f^{14}$ $5d^1$ $6s^2$

(3) [Kr] $5s^1$

(4) [Ar] $3d^7$ $4s^2$

17) 1L N_2 gas, $\frac{7}{8}$ L O_2 gas and 1L CO are taken in mixture under identical conditions of P &T. Amount of gases present in mixture is given by-

(1) $W_{N_2} = W_{O_2} > W_{CO}$

(2) $W_{N_2} = W_{CO} > W_{O_2}$

(3) $W_{N_2} = W_{O_2} = W_{CO}$

(4) $W_{CO} > W_{N_2} > W_{O_2}$

18) The mass of 11.2 L of any gas at STP represents

(1) gram molar mass

(2) gram equivalent mass

(3) gram atomic mass

(4) vapour density

19) Pair of species having same percentage weight of carbon are

(1) $C_6H_{12}O_6$ & $C_{12}H_{22}O_{11}$

(2) CH_3COOH & $C_6H_{12}O_6$

(3) CH_3COOH & C_2H_5OH

(4) $HCOOCH_3$ & $C_{12}H_{22}O_{11}$

20) Mass of oxygen required for combustion of 1 kg of a mixture of butane and isobutane is :

(1) 1.8 kg

(2) 2.7 kg

(3) 4.5 kg

(4) 3.58 kg

21) For reaction $C + O_2 \rightarrow CO_2$ carried out by 24 gm carbon & 96 gm O_2. Determine mass of excess reactant remaining.

(1) C, 12 gm

(2) C, 16 gm

(3) O_2, 32 gm

(4) O_2, 40 gm

22) Largest number of molecules are in

(1) 36 gm H_2O

(2) 28 gm CO

(3) 46 gm C_2H_5OH

(4) 54 gm N_2O_5

23) If a mixture of 3 moles of H_2 & 1 mole N_2 is completely converted into NH_3, ratio of initial & final total volumes under same temperature & pressure is :

(1) 3 : 1

(2) 1 : 3

(3) 2 : 1

(4) 1 : 2

24) The heats of neutralisation of four acids P, Q, R, S are –13.7, –9.4, –11.2, –12.4 kcal respectively when they are neutralised by a common base. Order of acidic character is :

(1) P > Q > R > S

(2) P > S > R > Q

(3) S > R > Q > P

(4) S > Q > R > P

25) Enthalpy of formation of $CO_{2(g)}$ & $CaO_{(s)}$ are – 94 kJ/mol & –152 kJ/mol respectively & enthalpy of reaction $CaCO_{3(s)} \rightarrow CaO_{(s)} + CO_{2(g)}$ is 42 kJ/mol. Enthalpy of formation of $CaCO_{3(s)}$ is :

(1) –42 kJ/mol

(2) –202 kJ/mol

(3) +202 kJ/mol

(4) –288 kJ/mol

26) $H_{2(g)} + \frac{1}{2}O_{2(g)} \rightarrow H_2O_{(\square)}$; ΔH_{298K} = –68.32 kcal
Heat of vapourization of water at 1 atm & 25°C is 10.52 kcal then standard heat of formation (in

kcal) of 1 mole of $H_2O_{(g)}$ at 25°C is :

(1) –78.84
(2) 78.84
(3) +57.80
(4) –57.80

27) Standard enthalpy of formation of CH_4, C_2H_4 & C_3H_8 are –17.9, 12.5, –24.8 kcal/mole respectively then ΔH for reaction is :
$CH_{4(g)} + C_2H_{4(g)} \rightarrow C_3H_{8(g)}$

(1) –30.2 kcal
(2) +30.2 kcal
(3) +19.4 kcal
(4) –19.4 kcal

28) When 2.1 gm Fe combines with S the heat released is 3.75 kJ. Then heat of formation of FeS in kJ/mol is :

(1) –3.77
(2) –1.79
(3) –100
(4) None of above

29) A gaseous reaction attains equilibrium $A_{(g)} \rightleftharpoons B_{(g)} + C_{(g)}$ at Constant pressure P. If partial pressure of A at equilibrium is $\frac{P}{2}$. Calculate K_P.

(1) $\frac{P}{2}$
(2) $\frac{P}{4}$
(3) $\frac{P}{6}$
(4) $\frac{P}{8}$

30) 20 gm of $CaCO_3$ in 10 L flask is heated to 819°C, 35% of $CaCO_3$ remained unreacted at equilibrium. Calculate K_P.
(Use R = $\frac{1}{12}$ L atm mol^{-1} K^{-1})

(1) 11.83 atm
(2) 21.83 atm
(3) 1.183 atm
(4) 3.183 atm

31) Calculate 2nd exitation energy of He^+ ion

(1) 12.09 eV

(2) 48.36 eV

(3) 12.75 eV

(4) 40 eV

32) Identify the correct option

(P)	Isotopes	(1)	A – 2Z = same
(Q)	Isodiaphors	(2)	No. of e^- = same
(R)	Isotones	(3)	Z = same
(S)	Iso electronic species	(4)	A – Z = same

(1) P→4 ; Q→1 ; R→3 ; S→4

(2) P→3 ; Q→4 ; R→4 ; S→3

(3) P→3 ; Q→1 ; R→4 ; S→2

(4) P→1 ; Q→2 ; R→1 ; S→1

33) Calculate the ratio of energy difference of 3rd & 2nd orbit of He^+ ion & 2nd & 1st orbit of Li^{+2} ion.

(1) $\frac{20}{243}$

(2) $\frac{5}{108}$

(3) $\frac{108}{5}$

(4) $\frac{243}{20}$

34) For an H-like atom 114.75 eV is the 3rd exitation energy then calculate Z of atom.

(1) Z = 1

(2) Z = 2

(3) Z = 3

(4) Z = 4

35) 50% neutralization of a solution of formic acid ($K_a = 2 \times 10^{-4}$) with NaOH would result in a solution having hydrogen ion concentration of

(1) 2×10^{-4}

(2) 3.7

(3) 2.7

(4) 1.85

36) Which would decrease the pH of 25 cm^3 of a 0.01 M solution of HCl.

(1) Addition of Mg metal

(2) Addition of 25 cm^3 of 0.02 M HCl

(3) Addition of 25 cm^3 of 0.005 M HCl

(4) None of above

37) Concentration of fluro acetic acid ($K_a = 2.7 \times 10^{-3}$) which is required to get $[H^+] = 1.5 \times 10^{-3}$ M is :

(1) 2.33×10^{-3} M

(2) 2.33×10^{-2} M

(3) 2.33×10^{-4} M

(4) 0.833 M

38) pH of a solution formed by mixing 40 mL of 0.1 M HCl & 10 mL of 0.45 M NaOH is ?

(1) 5

(2) 8

(3) 12

(4) 10

39) K_a of formic acid & acetic acid having same concentration are 2.2×10^{-4} & 1.1×10^{-5} respectively. Relative strength of acids is :

(1) 2 : 1

(2) 1.414 : 1

(3) 1 : 1.414

(4) 4.47 : 1

40) To 100 mL of 0.1 M $AgNO_3$ solution solid K_2SO_4 is added. At which concentration of K_2SO_4, the precipitation of Ag_2SO_4 will begin (K_{sp} of $Ag_2SO_4 = 6.4 \times 10^{-5}$)

(1) 6.4×10^{-8} M

(2) 6.4×10^{-3} M

(3) 6.4×10^{-7} M

(4) 6.4×10^{-5} M

41) pH of a solution containing 0.1 N NaOH & 0.1 N NH_4OH is :

(1) 1

(2) 7

(3) 2

(4) 13

42) With which element oxygen shows positive oxidation state in it's compounds

(1) Na

(2) Cl

(3) N

(4) F

43) Oxidation state of nitrogen in NH_4NO_3 is :

(1) +3

(2) +5

(3) +3 & +5

(4) –3 & +5

44) What mass of HNO_3 is needed to convert 5 gm of iodine into iodic acid according to the reaction $I_2 + HNO_3 \rightarrow HIO_3 + NO_2 + H_2O$

(1) 12.4 gm

(2) 24.8 gm

(3) 0.248 gm

(4) 49.6 gm

45) What volume of 0.1 N oxalic acid solution can be oxidsed by 250 g of a 8% by mass $KMnO_4$ solution ? (Atomic wt. K = 39, Mn = 55, O = 16)

(1) 6.3 L

(2) 12.6 L

(3) 25.2 L

(4) 0.63 L

BIOLOGY

1) Which of the following statement is **not** correct?

(1) The sum total of all the catabolic reations occurring in our body is metabolism.

(2) Metabolism is a defining feature of all living organism without exception.

(3) Cellular organisation of the body is the defining feature of life forms.

(4) All living organisms are made of chemicals.

2) The term 'Taxon' is used for :-

(1) The rank of species and genus

(2) The rank upto phylum

(3) The species epithet only

(4) Any rank of taxonomic hierachy

3) When taxonomy is based on chromosome number, structure and behaviour then it is called as :-

(1) Chemotaxonomy

(2) Numerical taxonomy

(3) Cytotaxonomy

(4) Systematics

4)

Which of following is not a universal rule of nomenclature ?

(1) Biological names are generally in Latin and printed in italics

(2) First word of a biological name represents the genus

(3) First word in a zoological name is specific epithet

(4) In *Mangifera indica* Linn., the word "Linn." indicates that the species was first described by Linnaeus

5) How many is/are the correct statements from the following?
(A) Each taxonomic category is a unit of classification
(B) Taxonomic groups are distinct biological entities
(C) Genera are aggregates of closely related families
(D) Both vegetative and reproductive characters are used to characterise the families.

(1) Two

(2) Three

(3) One

(4) Four

6) According to five kingdom classification multicellular eukaryotes are mainly classified on the basis of :-

(1) Phylogeny

(2) Mode of nutrition

(3) Mode of reproduction

(4) Body organisation

7) The structures that help some bacteria to attach to rocks in streams and also to the host tissues are :

(1) Flagella

(2) Cilia

(3) Mesosomes

(4) Fimbriae

8) Which of the following combination of characters is **true** for slime moulds?

(1) Parasitic, plasmodium without walls, spores dispersed by air currents

(2) Saprophytic, plasmodium with walls, spores dispersed by water

(3) Parasitic, plasmodium without walls, spores dispersed by water

(4) Saprophytic, plasmodium without walls, spores dispersed by air currents

9) In which of the following plasmogamy is followed by karyogamy immediately?

(1) *Mucor*
(2) *Aspergillus*
(3) *Ustilago*
(4) *Puccinia*

10) **Statement I :** The viruses are non-cellular organisms that are characterised by having an inert crystalline structure outside the living cell.
Statement II : The RNA of the viroid is of high molecular weight.

(1) Statement-I is incorrect but Statement-II is correct.
(2) Statement-I is correct but Statement-II is incorrect.
(3) Both Statement-I and Statement-II are correct.
(4) Both Statement-I and Statement-II are incorrect.

11) Match List-I with List-II :

	List-I		**List-II**
(a)	*Neurospora*	(i)	Citric acid
(b)	Yeast	(ii)	Antibiotics
(c)	*Aspergillus*	(iii)	Bread and beer
(d)	*Penicillium*	(iv)	Bio-chemical and genetic work

Choose the correct answer from the options given below :
(1) (a)-(i), (b)-(iii), (c)-(ii), (d)-(iv)
(2) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)
(3) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)
(4) (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)

12) Given below are two statements :- one is labelled as assertion (A) and other is labelled as reason (R) :-
Assertion (A) : According to five kingdom classification *Euglena*, *Amoeba* and *Paramoecium* are included in kingdom protista.
Reason (R) : They all are unicellular eukaryotes.
In the light of above statements, choose correct answer from the options given below :-

(1) Both (A) and (R) are **true** and (R) is the **correct explanation** of (A).
(2) Both (A) and (R) are **true** and (R) is the **not correct explanation** of (A).
(3) (A) is **true** but (R) is **false**.
(4) (A) is **false** but (R) is **true**.

13) **Assertion (A) :** Lichens are symbiotic association between algae and fungi. The algal

component it known as phycobiont, which is autotrophic and fungal component is known as mycobiont which is heterotrophic.
Reason (R) : Lichens are very good pollution indicators they do not grow in polluted areas.
In the light of the above statements, choose the correct answer from the options given below :-

(1) Both A and R are true and R is correct explanation of A.

(2) Both A and R are true and R is not correct explanation of A.

(3) A is true but R is false.

(4) A is false but R is true.

14) Among dinoflagellates, cyanobacteria, mycoplasma, desmids, diatoms, euglenoids, slime moulds, archaebacteria how many are included in kingdom Monera according to R.H. Whittaker's five kingdom system of classification ?

(1) Four

(2) Three

(3) two

(4) Six

15) Read the following statements :-
(i) Diatoms are the chief producers in the oceans.
(ii) Slime moulds are photosynthetic protists.
(iii) Ciliated protozoans have a cavity that opens to the outside of the cell surface.
(iv) All protozoans are saprophytic protist and live as predators
How many of the above statements is/are correct ?

(1) 3

(2) 1

(3) 2

(4) 4

16) Which one of the following shows isogamy with non-flagellated gametes?

(1) *Sargassum*

(2) *Ectocarpus*

(3) *Ulothrix*

(4) *Spirogyra*

17) Which of the following is **not** correct regarding importance of algae?

(1) At least half of the total carbon dioxide fixation on earth is carried out by algae through photosynthesis.

(2) Being photosynthetic they increase the level of dissolved oxygen in their immediate environment.

(3) Many species of *Porphyra*, *Laminaria* and *Sargassum* are among the 70 species of marine algae used as food.

(4) *Gracilaria* a unicellular alga rich in proteins is used as food supplement even by space travellers.

18) The predominant stage in the life cycle of Moss is gametophyte, which consists of two stages they are respectively:-

(1) Protonema, Prothallus

(2) Spore mother cell, protonema

(3) Protonema, leafy stage

(4) Protonema, zygote

19) **Assertion :** In Pteridophytes sporangia produce spores by meiosis in spore mother cell.
Reason : In Pteridophytes, spore germinate in soil and produce multicellular sporophytes.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

20) In gymnosperms microspores and megaspores are produced by the process of respectively:

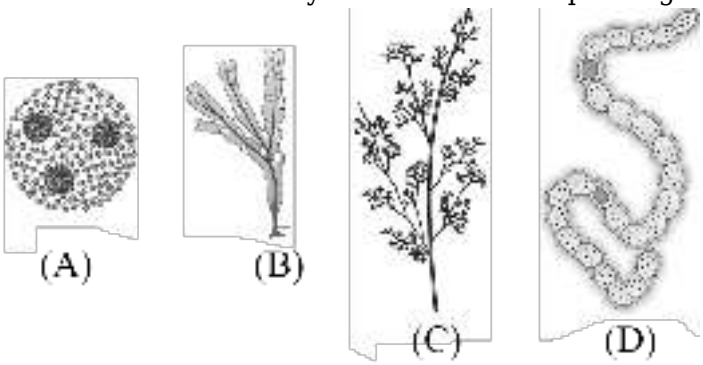
(1) Meiosis and Meiosis

(2) Meiosis and Mitosis

(3) Meiosis and Budding

(4) Meiosis and Fragmentation

21) The four sketches (A, B, C and D) given below represent four different types of organism. Which one of these is correctly identified in the options given, along with its correct stored food and class?



		Name of Organism	**Stored Food**	**Class**
(1)	(D)	*Nostoc*	Glycogen	Chlorophyceae
(2)	(A)	*Volvox*	Floridean starch	Chlorophyceae
(3)	(B)	*Laminaria*	Starch	Phaeophyceae
(4)	(C)	*Polysiphonia*	Floridean starch	Rhodophyceae

(1) 1

(2) 2

(3) 3

(4) 4

22) Consider the following four statements (a-d) and select the option which include all the correct ones only :-
(a) Unlike bryophytes and pteridophytes, in gymnosperms the male and female gametophyte do not have an independent free living existence.
(b) In some pteridophytes, the female gametophyte is retained on the parent sporophyte for variable period.
(c) In pteridophytes spores are produced from spore mother cell by meiosis.
(d) Bryophytes do not play any role in plant succesion occuring on bare rocks or soil or terrestrial habitats.

(1) b, c and d

(2) a, c and d

(3) a, b and d

(4) a, b and c

23) Few stages of life cycle of moss are given below. Arrange these events in correct order starting from sex organ bearing structure and select the correct option from option given below :
A. Protonema
B. Leafy gametophyte
C. Spore formation
D. Spore germination
E. Fertilization
F. Sporophyte

(1) A → B → C → D → F → E

(2) A → B → C → D → E → F

(3) A → B → F → E → C → D

(4) B → E → F → C → D → A

24)

Which of the following are not correctly matched?

A	Packaging of macromolecules	Golgi body
B	Catalase enzyme	SER
C	Connect cytoplasm of adjacent animal cells	Plasmodesmata
D	Detoxification of drugs & carcinogen	SER
E	Anthocyanin pigment	Chromoplast
F	Protein factory of cell	Nucleolus

(1) B, D, E, F

(2) A, C, D, E

(3) A, B, C, E

(4) B, C, E, F

25) Read the following statements carefully :
(A) Cholesterol is present in cell membrane
(B) Polar molecules can pass through the lipid bilayer of plasma membrane, therefore they do not require carrier proteins to facilitate their transport.
(C) Secondary wall is capable of growth and it is formed on the outer side of the cell.
(D) Quasifluid nature of lipid enables lateral movement of proteins within the overall lipid bilayer of plasma membrane.
(E) Middle lamella glues the different neighbouring plant cells together.
How many statements are incorrect ?

(1) Three
(2) Five
(3) Four
(4) Two

26) **Assertion :-** Some parts of plant appears orange, yellow or red in colour.
Reason :- Chromoplast present in coloured plant parts contains fat soluble carotenoid pigments like carotene, xanthophyll etc.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

27) Match **List - I** with **List - II.**

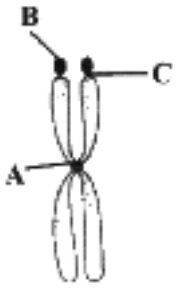
List - I		**List - II**	
(a)	Chloroplast	(i)	Proteins
(b)	Amyloplasts	(ii)	Oil and fats
(c)	Elaioplasts	(iii)	Starch
(d)	Aleuroplasts	(iv)	Chlorophyll

Choose the **correct answer** from the options given below :
(1) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)
(2) (a)-(iv), (b)-(ii), (c)-(iii), (d)-(i)
(3) (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)
(4) (a)-(iv), (b)-(i), (c)-(iii), (d)-(ii)

28) Which of the following structures are not present in the structure of a chromosome ?
A. Chromatid B. Centrosome
C. Satellite D. Kinetochore
E. Axoneme

(1) A, B, C & D
(2) A, C & D
(3) B & E

(4) Only E



29) What does A, B and C represent in the given figure of a chromosome ?

	A	B	C
(1)	Centriole	Satellite	Primary constriction
(2)	Centriole	Satellite	Secondary constriction
(3)	Centromere	Satellite	Secondary constriction
(4)	Centromere	Satellite	Primary constriction

(1) 1
(2) 2
(3) 3
(4) 4

30) Conversion of H_2O_2 into H_2O and O_2 occurs in peroxisome by the enzyme _____

(1) Lipase
(2) Catalase
(3) Urease
(4) Amylase

31) Hydrolases - lipases, proteases and carbohydrases in animal cells are found in :-

(1) Lysosome
(2) Vacuole
(3) Endoplasmic reticulum
(4) Chloroplast

32) Which cell organlles is involved in ormosegulation ?

(1) Mitochondria
(2) Chloroplast
(3) ER
(4) Contractile vacoule

33) Match the column-I with column-II and select correct answer from given options.

Column-I	Column-II

(A)	Certriole duplication	(I)	G_0
(B)	Tubulin synthesis	(II)	G_1
(C)	Duplication of cell orgenells like ER, Golgi complex, mitochondna etc.	(III)	S
(D)	Exit from G_1 stage	(IV)	G_2

(1) A-IV, B-III, C-II, D-I

(2) A-I, B-II, C-III, D-IV

(3) A-III, B-IV, C-II, D-I

(4) A-IV, B-III, C-I, D-II

34) Match the Column-I with Column-II and select the correct option:-

Column-I		**Column-II**	
(A)	Prophase	(I)	Chromosomes cluster at opposite spindle poles and their identify is lost as discrete element.
(B)	Metaphase	(II)	Spindle fibre attaches to kinetochore of chromosome
(C)	Anaphase	(III)	Initiation of condensation of chromosomal material
(D)	Telophase	(IV)	Centromere split and chromatids separate

(1) A-IV, B-II, C-I, D-III

(2) A-III, B-II, C-IV, D-I

(3) A-I, B-III, C-II, D-IV

(4) A-I, B-II, C-IV, D-III

35) Which of the following statement is incorrect about Meiosis-I?

(1) During zygotene stage chromosomes starts pairing together.

(2) In Oocytes of some vertebrates, diplotene can last for months or years.

(3) Crossing over occurs between the sister chromatids of homologous chromosomes.

(4) Diakinesis represents transition to metaphase I

36) In a diploid cell before s-phase quantity of DNA is 40 pico gram (pg) and chromosome number is 40 then what will be the amount of DNA and chromosome number in prophase-I, anaphase-I, after meiosis-I and after meiosis-II ?

	Prophase-I	**Anaphase-I**	**After meiosis-I**	**After meiosis-II**
(1)	40 pg, 40	40 pg, 40	20 pg, 40	20 pg, 20
(2)	80 pg, 40	40 pg, 40	20 pg, 20	10 pg, 10
(3)	40 pg, 40	20 pg, 40	20 pg, 20	10 pg, 10
(4)	80 pg, 40	80 pg, 40	40 pg, 20	20 pg, 20

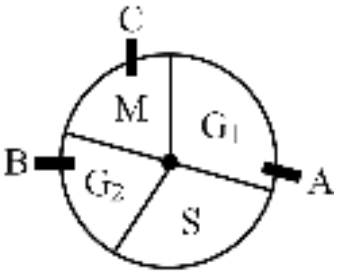
(1) 1

(2) 2

(3) 3

(4) 4

37) Identify C in the given figure of cell cycle -



(1) G_1-S transition

(2) G_2-M transition

(3) APC

(4) G_0 phase

38) Mitotic apparatus is :

(1) Centrosome radiates out microtubules only

(2) Chromosome attached with spindle fibres

(3) Two centrosomes attached with microfilaments

(4) Two asters together with spindle fibres

39) Identify the given figure and choose correct answer :



(1) Homologous chromosomes separation

(2) Splitting of the centromere of each chromosome

(3) The bivalent chromosomes align on the equatorial plate

(4) Chromosomes get enclosed by a nuclear envelope

40) **Statement-I :** Fatty acids could be saturated (without double bond) or unsaturated (with one or more C = C double bonds).
Statement-II : Some tissues specially the neural tissues have lipids with more complex structures.
Select correct option for given above statements.

(1) Statement-I is correct and Statement-II are incorrect.

(2) Statement-I is incorrect and Statement-II are correct.

(3) Both Statement-I and Statement-II are incorrect.

(4) Both Statement-I and Statement-II are correct.

41) Match the following :-

	Column-I		Column-II
(a)	GLUT-4	(i)	Fights infectious agents
(b)	Antibody	(ii)	Enables glucose transport into cells
(c)	RuBiSCo	(iii)	Most abundant protein on the earth

(1) a-i, b-ii, c-iii

(2) a-iii, b-i, c-ii

(3) a-ii, b-i, c-iii

(4) a-ii, b-iii, c-ii

42) The starch-I_2 is _____ in colour. Cellulose does not contain complex_____ and hence cannot hold I_2 :
Fill in the blanks, respectively :-

(1) Red; Helices

(2) Blue; Helices

(3) Red; Straight chain

(4) Blue; Branches

43) Which bond is hydrolyzed to convert a disaccharide into monosaccharides ?

(1) Hydrogen bond

(2) Glycosidic bond

(3) Phosphodiester bond

(4) Disulfide bond

44) Match the column-I with column-II and select the correct answer.

	Column-I		Column-II
(a)	Oxidoreductase	(i)	Enzymes which catalyse oxidoreduction
(b)	Transferase	(ii)	Enzymes catalysing hydrolysis
(c)	Ligases	(iii)	Enzymes cataysing a transfer of a group
(d)	Hydrolases	(iv)	Enzymes catalysing the linking together of 2 compounds.

(1) a-i, b-ii, c-iii, d-iv

(2) a-ii, b-i, c-iv, d-iii

(3) a-iii, b-ii, c-i, d-iv

(4) a-i, b-iii, c-iv, d-ii

45) Substrate binds at which site of enzyme ?

(1) Allosteric site
(2) Inhibitory site
(3) Active site
(4) Promotery site

46) Sea lily is the common name of :

(1) *Asterias*
(2) *Branchiostoma*
(3) *Doliolum*
(4) *Antedon*

47) Which sponge is found in the river ?

(1) *Sycon*
(2) *Spongilla*
(3) *Meandrina*
(4) (2) & (3) both

48) To regulate buoyancy and some times it also function like lungs in bony fishes, this organ is :

(1) Pneumatic bones
(2) Air bladder
(3) Electric organ
(4) Operculum

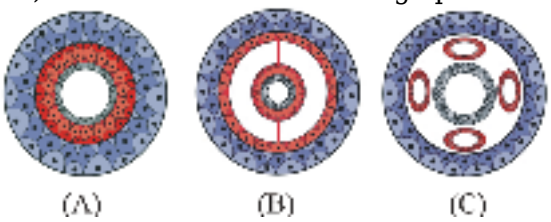
49) Member of class reptilia but have 4 chambered heart–

(1) Crocodile
(2) Snake
(3) Garden lizard
(4) Tree lizard

50) **Assertion (A) :-** Hemichordata was earlier considered as a phylum under non-chordata. But now is placed as a sub-phylum of chordata.
Reason (R) :- Hemichordata have a rudimentary structure in collar region called stomochord.

(1) A and R are correct but R, is not correct explaination of A
(2) A and R are correct and R, is correct explaination of A
(3) A is correct and R is incorrect
(4) A is incorrect and R is correct

51) Which one of the following option is correct about the figures A, B and C given below?



(1) A-Acoelomate, B-Pseudocoelomate, C-Coelomate

(2) A-Coelomate, B-Acoelomate, C-Pseudocoelomate

(3) A-Pseudocoelomate, B-Coelomate, C-Acoelomate

(4) A-Acoelomate, B-Coelomate, C-Pseudocoelomate

52) Match the column-I with column-II :-

Column-I		**Column-II**	
(i)	Choanocytes	A	Roundworm
(ii)	Bioluminescence	B	Sycon
(iii)	Proboscis gland	C	Pleurobrachia
(iv)	Well-developed muscular pharynx	D	Balanoglaussus

(1) (i)-A, (ii)-B, (iii)-C, (iv)-D

(2) (i)-D, (ii)-B, (iii)-A, (iv)-C

(3) (i)-C, (ii)-D, (iii)-A, (iv)-B

(4) (i)-B, (ii)-C, (iii)-D, (iv)-A

53) **Statement-I :** In cyclostomates anadromous migration occurs for the purpose of spawning.
Statement-II : After spawning, cyclostomates return to their original habitat i.e. Marine habitat.

(1) Both **Statement I** and **Statement II** are incorrect.

(2) **Statement I** is correct but **Statement II** is incorrect.

(3) **Statement I** is incorrect but **Statement II** is correct

(4) Both **Statement I** and **Statement II** are correct.

54) Match the following :

(A)	Agnatha	(1)	*Lamprey*
		(2)	Hag fish
(B)	Gnathostomata	(3)	Sting Ray
		(4)	Saw fish
		(5)	Katla

(1) A-3,4,5 ; B-1,2

(2) A-1,2,3 ; B-3,4

(3) A-1,2, ; B-3,4,5

(4) A-2,3,4 ; B-2,4,5

55) Read the following statements and choose the correct statements :- (a) Hemichordata was earlier considered as a sub-phylum under phylum chordata.
(b) hemichordates have a rudimentary structure in the collar region called stomochord.
(c) Structure of stomochord is similar to notochord.
(d) The most distinctive feature of echinoderms is the presence of water transport system.
(e) The adult echinoderms are radially symmetrical but larvae are bilaterally symmetrical.

(1) (a), (b), (c) & (e)

(2) Only (c)

(3) (b), (c) & (d)

(4) (a), (b), (c), (d) & (e)

56) Select the incorrect statement regarding cockroach ?

(1) Head is triangular in shape.

(2) In each segment of cockroach exoskeleton has hardened plates called sclerites

(3) The head is connect with abdomen.

(4) Forewings are present on mesothorax

57) Match list-I with list-II.

	List-I (Cockroach)		**List-II (Term used)**
A.	Sclerites	I.	Mosaic vision
B.	Supra-oesophageal ganglion	II.	Sound receptor
C.	Ommatidia	III.	Brain
D.	Anal cerci	IV.	Exoskeleton

Choose the correct answer from the options given below.

(1) A-II, B-III, C-I, D-IV

(2) A-IV, B-III, C-I, D-II

(3) A-III, B-IV, C-II, D-I

(4) A-IV, B-I, C-III, D-II

58) In cockroach Sclerite joined to each other by -

(1) Thin and flexible articular membrane

(2) Thin and non-flexible Arthroidal memebrane

(3) Thick and flexible Arthroidal membrane

(4) Thick and non-flexible Articular membrane

59) Match the column A with coumn B and select the correct option.

Column A		Column B	
(a)	Skin of Frog	(i)	Ventral surface
(b)	Heart of Frog	(ii)	Closed type
(c)	Blood circulation of Frog	(iii)	Incomplete double
		(iv)	Moist and Glandular
		(v)	Three chambered
		(vi)	Help in respiration

(1) a-(iv), (i) ; b-(ii), (v) ; c-(iii), (vi)
(2) a-(iv), (vi) ; b-(i), (v) ; c-(ii), (iii)
(3) a-(iv), (vi) ; b-(ii), (iii) ; c-(i), (v)
(4) a-(v), (vi) ; b-(iii), (i) ; c-(ii), (iv)

60) The cloaca is a _A_ chamber that is used to pass _B_,_C_ and _D_ to the exterior.

(1) A–small median, B–faeces, C–urine, D–Ovum
(2) A–Large median, B–faeces, C–urine, D–Sperm
(3) A–small median, B–faeces, C–urine, D–Sperm
(4) A–small median, B–faeces, C–urine, D–Sperm or ovum

61) Find the incorrect statement regarding frog.

(1) The vascular system of frog is well-developed closed type
(2) The excretory system consists of a pair of kidney, ureters a cloaca and urinary bladder.
(3) In female ureters acts as urinogenital duct which opens into cloaca
(4) The frog excretes urea and thus is a ureotelic animal

62) In frog a triangular structure called sinus venosus joins which chamber of heart ?

(1) Right ventricle
(2) Left ventricle
(3) Right atrium
(4) Left atrium

63) Frog liver secrete bile that is stored in :-

(1) Pancreas
(2) Cloaca
(3) Gall-Bladder
(4) Duodenum

64) Which of the following tissue are most abundant and widely distributed in the body of complex animals :-

(1) Epithelial
(2) Connective
(3) Muscular
(4) Nervous

65) Which epithelial tissue present in the lining of stomach and intestine ?

(1) Simple columnar epithelium
(2) Simple cuboidal epithelium
(3) Simple squamous epithelium
(4) Compound epithelium

66) Match the correct one-

(1) Simple columner epithelium → Alveoli
(2) Simple cuboidal epithelium → Stomach
(3) Simple columner epithelium → Stomach
(4) Simple squamous epithelium → Intestine

67) Match column-I with column-II and select the correct option-

Column-I		**Column-II**	
a	Areolar tissue	i	Packing between organs
b	Adipose tissue	ii	Tendons and ligaments
c	Dense regular tissue	iii	Long term fat storage
		iv	Most abundant type of connective tissue
		v	Insulation

(1) a-iv; b-ii, v; c-i, iii
(2) a-ii, iv; b-iii, v; c-i, v
(3) a-i, iii; b-ii,iv; c-v
(4) a-i, iv; b-iii, v; c-ii

68) **Statement-I :** Ventricular systole increase the ventricular pressure causing the closure of tricuspid & Bicuspid values due to attempted backflow of blood into atria. **Statement-II :** During relaxation of ventricles, the ventricular pressure decreases causing closure of semilunar values. Which prevent back flow of blood into ventricles.
What will not happens if semilunar valve fails to close completely ?

(1) Backflow of blood into ventricles
(2) Increase end systolic volume
(3) Decrease cardiac output
(4) Increase systolic pressure

69) Match the following-

Column-I		Column-II		Column-III	
A.	Basophils	P.	20-25%	i.	Inflammation
B.	Lymphocytes	Q.	0.5-1%	ii.	Phagocytosis
C.	Neutrophils	R.	2-3%	iii.	Allergy
D.	Eosinophils	S.	60-65%	iv.	Cell Mediated immunity

(1) A-S-iii, B-Q-i, C-R-iv, D-P-ii

(2) A-P-iv, B-Q-i, C-R-ii, D-S-iiI

(3) A-Q-i, B-P-iv, C-S-ii, D-R-iii

(4) A-Q-i, B-S-ii, C-P-iv, D-R-iii

70) **Statement-I:** The QRS complex represents the depolarisation of the ventricles.
Statement-II: The T-wave represents the return of the ventricles from excited to normal state & the end of the T-wave marks the beginning of joint diastole.

(1) Both statement are correct

(2) Both statement are incorrect

(3) Statement-I is correct but statement-II is incorrect

(4) Statement-I is incorrect but statement-II is correct.

71) A patient feels discomfort only while climbing stairs or during increased physical activity. The pain in chest region lasts 1-2 minutes and disappears after rest. No permanent heart muscle damaged is observed. The condition is :

(1) Cardiac arrest

(2) Angina

(3) Myocardial infarction

(4) Heart failure

72) In the ECG of a person QRS wave is not normal, then it may indicates -

(1) Complete heart is not working properly.

(2) Atrial depolarisation is abnormal

(3) Atrial repolarisation in abnormal

(4) Ventricular depolarisation in abnormal

73) Choose an agranulocyte which is phagocytic in nature ?

(1) Neutrophils

(2) Monocytes

(3) Both 1 & 2

(4) Eosinophils

74) In humans the atrium and the ventricle of the same side are also separated by _____ tissue called the atrio- ventricular septum.

(1) Thick muscular

(2) Thin muscular

(3) Thin fibrous

(4) Thick fibrous

75) Ventricle of amphibians and reptiles pump which type of blood?

(1) Only oxygenated blood

(2) Only deoxygenated blood

(3) Oxygenated and deoxygenated blood separately

(4) Mixed blood

76) Given below are two statements :
Statement-I : The diffusion membrane is made up of three major layers namely, the thin squamous epithelium of alveoli, the endothelium of alveolar capillaries and the basement substance in between them, However its total thickness is much less than a micrometre.
Statement-II : Alveoli are the primary sites of exchange of gases.
In the light of the above statements, choose the most appropriate answer from the options given below :

(1) Both statement-I and statement-II are correct.

(2) Both statement-I and statement-II are incorrect.

(3) Statement-I is correct but statement-II is incorrect.

(4) Statement-I is incorrect but statement-II is correct.

77) Match **List I** with **List II**:

	List I		**List II**
A.	sponges, Coelenterates and flatworms	i.	Respiration through gills
B.	Earthworms	ii.	Respiration through general body surface
C.	Insects	iii.	Respiration through moist cuticle
D.	Aquatic Arthropods and Molluscus	iv.	Respiration through tracheal tubes

Select correct answer from the options given below:

(1) A-i, B-iv, C-iii, D-ii

(2) A-iv, B-iii, C-ii, D-i

(3) A-ii, B-iii, C-iv, D-i

(4) A-iii, B-ii, C-i, D-iv

78) Match the column-I with column-II and select the correct options

	Column I		Column II
(a)	PO_2 in Alveoli	(i)	sensitive for CO_2 and H^+
(b)	High Temp.	(ii)	Breathing
(c)	Carotid body	(iii)	104 mmHg.
(d)	I^{st} step of Respiration	(iv)	Right shift in O_2 dissociation curve

(1) a-i, b-ii, c-iii, d-iv

(2) a-iv, b-iii, c-ii, d-i

(3) a-iii, b-iv, c-i, d-ii

(4) a-ii, b-iv, c-iii, d-i

79) **Assertion :** Oxygen is utilised by the organism to indirectly break down the simple molecule like glucose, amino acids and fatty acids.
Reason : Exchange part of is the site of actual diffusion of O_2 and CO_2 between blood and atmospheric air.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion

(2) Both Assertion and Reason are true and Reason is not the correct explanation of Assertion

(3) Assertion is correct but Reason is incorrect

(4) Assertion is incorrect but Reason is correct.

80) Exchange part of respiratory system is made up of :

(1) From external nostrils upto the primary bronchi

(2) from external nostrils upto the Alveoli

(3) from trachea to alveoli

(4) Alveoli and their ducts

81) Contraction in diagram will -

(1) Decrease the thoracic volume in antero - Posterior axis

(2) Decrease the thoracic volume at Dorso - ventral axis

(3) Increase the thoracic volume at Antero - Posterior axis

(4) Increasing the thoracic volume at Dorso - Ventral axis

82) Which of the following conditions are not favourable for dissociation of oxygen from oxyhaemoglobin in tissue ?

(1) Low pO_2

(2) High pH

(3) High H^+ concentration

(4) High temperature

83) Match column-I with column-II and select the correct option-

Column-I		Column-II	
a	Ammoniotelic animals	i	Cartilagenous fish
b	Ureotelic animals	ii	Insects
c	Uricotelic animals	iii	Frog
		iv	Reptiles
		v	Bony fish
		vi	Coelenterates

Select correct answer from the options given :-

(1) a-ii, iii; b-iv, v; (c)-i, vi

(2) a-i v; b-iii, vi; (c)-ii, iv

(3) a-v, vi; b-i, iii; (c)-ii, iv

(4) a-ii, v; b-iii, i; (c)-iv, vi

84) The first step in urine formation is the filtration of blood which is carried out by the glomerulus and called glomerular filtration. The glomerular capillary blood pressure causes filtration of blood, through 3 layers i.e. the endothelium of glomerular blood vessels, the epithelium of bowman's capsule and basement membrane between these two layers. Blood is filtered so finely through these membrane that almost all the constituents of plasma except proteins pass onto the lumen of Bowman's capsule.
Arrange the steps of glomerular filtration.
A → Glomerular capillary blood pressure is generated
B → Plasma minus proteins passes into Bowman's capsule.
C → Blood enters through afferent arteriole
D → Blood gets filtered through pores of endothelial cells.

(1) A → C → B → D

(2) C → A → D → B

(3) B → C → A → D

(4) C → B → D → A

85)

Match column-I with column-II

	Column-I		Column-II
(A)	Lungs	(i)	Sebum
(B)	Liver	(ii)	Bile
(C)	Sweat gland	(iii)	H_2O, Urea, Lactic acid, NaCl

(D)	Sebaceous gland	(iv)	CO_2

(1) (A) - i, (B) - ii, (C) - iii, (D) - iv

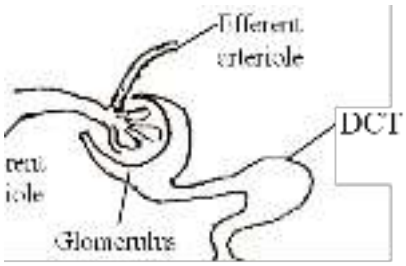
(2) (A) - ii, (B) - iv, (C) - iii, (D) - i

(3) (A) - iv, (B) - ii, (C) - i, (D) - iii

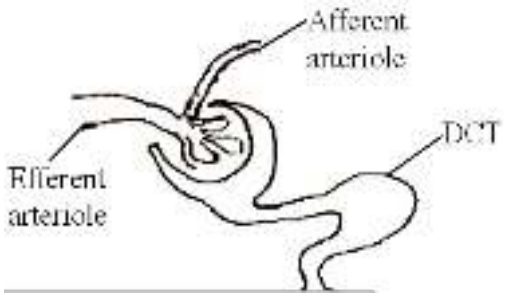
(4) (A) - iv, (B) - ii, (C) - iii, (D) - i

86) Which of the following diagrams is correct with regard to the Nephron ?

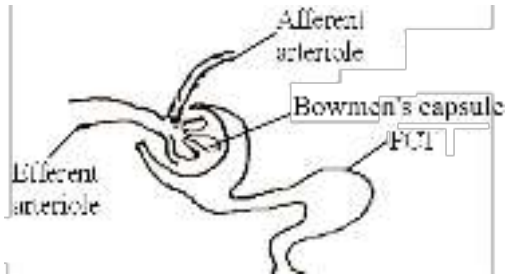
(1)



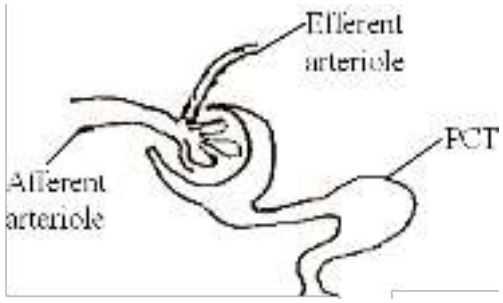
(2)



(3)



(4)



87) Which of the following is/are correct about counter current mechanism ?
(I) It maintains hyperosmolarity of medullary interstitium.
(II) It is contributed by both Henle's loop and vasa recta.
(III) Ascending limb of Henle's loop is permeable to water.
(IV) Urea contributes to interstitial hyperosmolarity.

(1) I, II and IV are correct

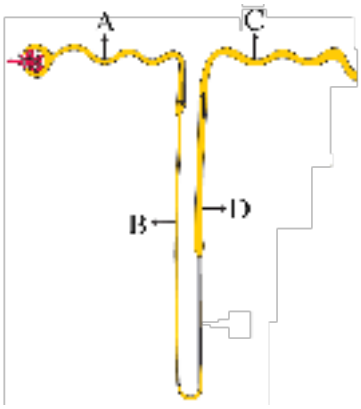
(2) II, III and IV are correct

(3) I, III and IV are correct

(4) All are correct

88)

Which parts of renal tubule labelled in the diagram shows minimum reabsorption -



(1) C

(2) D

(3) A

(4) B

89) A fall in GFR can activate the JG cells to release ___A___ which can stimulate the glomerular blood flow and thereby GFR back to normal. Fill in the blank.

(1) ADH

(2) Renin

(3) ANF

(4) Vasopressin

90) What is the main function of the juxta glomerular apparatus (JGA) in the human kidney ?

(1) Reabsorption of glucose

(2) Regulation of blood pressure & GFR

(3) Secretion of bile

(4) Formation of WBC

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	3	2	3	1	3	4	3	3	3	1	1	1	4	1	2	2	3	1	4	1
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	1	2	3	3	2	1	2	4	2	4	4	1	2	3	1	3	2	1	2	4
Q.	41	42	43	44	45															
A.	2	1	4	4	1															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	3	3	3	4	4	4	1	2	2	4	1	3	3	4	3	3	3	4	2	4
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	3	1	3	2	4	4	4	3	4	3	2	3	1	3	1	2	1	3	4	2
Q.	86	87	88	89	90															
A.	4	4	4	1	1															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	1	4	3	3	2	2	4	4	1	2	4	1	2	2	3	4	4	3	3	1
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	4	4	4	4	4	1	1	3	3	2	1	4	3	2	3	4	3	4	3	4
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	3	2	2	4	3	4	2	2	1	4	4	4	2	3	1	3	2	1	2	4
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	3	3	3	2	1	3	4	4	3	1	2	4	2	4	4	4	3	3	2	4
Q.	171	172	173	174	175	176	177	178	179	180										
A.	3	2	3	2	4	4	1	2	2	2										

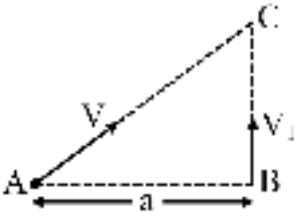
SOLUTIONS

PHYSICS

1) $$< v > = \frac{\text{Displacement}}{\text{Time}} = \frac{2R\sin\left(\frac{90^\circ}{2}\right)}{(\text{dist./speed})}$$

$$= \frac{\sqrt{2}R}{\frac{\pi R}{2v}} = \frac{2\sqrt{2}v}{\pi}$$

2) Let two boys meet at pt C after time t from starting then AC = Vt, BC = V_1t



$(AC)^2 = (AB)^2 + (BC)^2$

$(Vt)^2 = a^2 + V_1^2t^2$

$V^2t^2 = a^2 + V_1^2t^2$

$V^2t^2 - V_1^2t^2 = a^2$

$t^2[V^2-V_1^2] = a^2$

$$t = \sqrt{\frac{a^2}{V^2 - V_1^2}}$$

3) Let v_m be the maximum speed,

$$\left(\frac{v_m + 0}{2}\right) t_1 = \ell \Rightarrow t_1 = \frac{2\ell}{v_m}$$

$v_m \times t_2 = 2\ell \Rightarrow t_2 = \frac{2\ell}{v_m}$

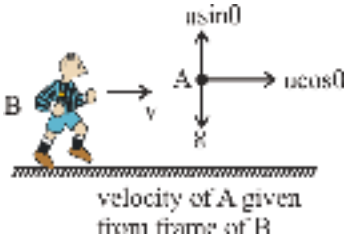
$$\frac{(v_m + 0)}{2} \times t_3 = 3\ell \Rightarrow t_3 = \frac{6\ell}{v_m}$$

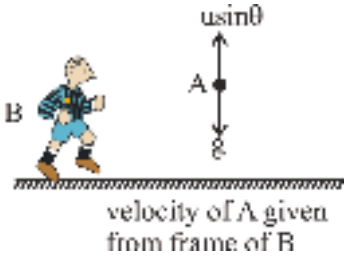
$$v_{avg.} = \frac{\ell + 2\ell + 3\ell}{t_1 + t_2 + t_3} = \frac{6\ell}{\frac{2\ell}{v_m} + \frac{2\ell}{v_m} + \frac{6\ell}{v_m}}$$

$$= \frac{6v_m}{10} \Rightarrow \frac{v_{avg.}}{v_m} = \frac{3}{5}$$

4) The horizontal and vertical components of initial velocity of projectile are as shown in figure. Since the observer moving with uniform velocity v sees the projectile moving in straight line

Hence v = u cos θ





The time of flight as measured by observer B

$$T = \frac{2u\sin\theta}{g}$$

Hence horizontal range of projectile on ground

$R = (u\cos\theta)T = vT$

5) Distance travelled by first train,

$$s_1 = \frac{u^2}{2a} = \frac{(15)^2}{2} = 112.5\text{m}$$

Distance travelled by second train,

$$s_2 = \frac{(20)^2}{2} = 200\text{m}$$

Total distance travelled = 312.5 m

Distance of separation = 500–312.5 = 187.5 m

6)

KE per unit mass = $\frac{V^2}{2}$ dimension (L^2T^{-2})

$$\frac{V^2}{2} = \alpha\gamma^x\sigma^y r^z$$

$$L^2T^{-2} = (ML^{-3})^x(M^1L^{-1}T^{-2})^yL^z$$

$$M^0L^2T^{-2} = M^{x+y}L^{-3x-y+z}T^{-2y}$$

$x + y = 0$ $\quad -3x - y + z = 2$ $\quad -2 = -2y$

$y = 1$ $\quad x = -1$ $\quad 3 - 1 + z = 2$

$z = 0$

$x + y + z = 0$

7)

The observed reading is the sum of the main scale reading and the product of the vernier scale division coincidence and the least count. The corrected reading is the observed reading minus the zero error. The mean corrected diameter is the average of the corrected readings.

Average $v_s = 6$

Reading $= 0.50 + 0.01 \times 6$

Corrected reading $= 0.56 + (0.03) = 0.59$ cm
The mean corrected diameter is 0.59 cm.

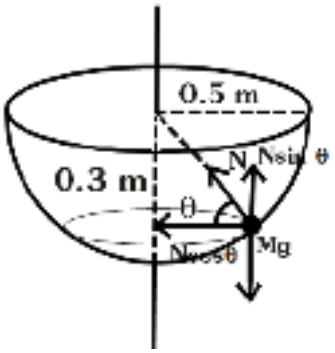
8) Answer since the acceleration of the box is due to the static friction.
$ma = f_s \leqslant \mu_s N = \mu_s mg$
i.e. $a \leqslant \mu_s g$
$\therefore a_{max} = \mu_s g = 0.15 \times 10 ms^{-2}$
$= 1.5\ ms^{-2}$

9) $T = \frac{F(L-x)}{L} = \frac{5(5-1)}{5} = 4N$

10) By using C.O.M.E.
$\frac{1}{2}(M+m)v^2 = (M+m)gh$
$v = \sqrt{2gh}$
By using C.O.L.M.
$mv' = (M + m)V$
$V' = \left(\frac{M+m}{m}\right)\sqrt{2gh}$

11)

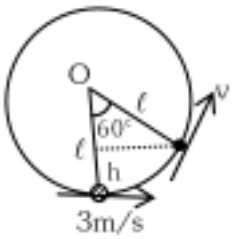
COM of uniform rod of length L $\rightarrow \frac{L}{2}$ from one end
COM of solid cone $\rightarrow \frac{H}{4}$ from base
COM of solid hemi sphere $\rightarrow \frac{3R}{8}$ from centre
COM of semi circular ring $\rightarrow \frac{2R}{\pi}$ from cetnre



12)
$N \sin\theta = mg$
$N \cos\theta = m\omega^2 r$
$\tan\theta = \frac{g}{\omega^2 r}$
$\omega = \sqrt{\frac{g}{r\tan\theta}}$

$$\omega = \sqrt{\frac{10}{0.4 \times \frac{0.3}{0.4}}}$$

$$\omega = \sqrt{\frac{100}{3}} = \sqrt{33.33} = 5.7 \text{ rad/sec}$$



13)

here $h = \ell - \ell \cos 60° = \frac{\ell}{2}$

COME

$k_1 + u_1 = k_2 + u_2$

$$\frac{1}{2}mu^2 + 0 = \frac{1}{2}mv^2 + \frac{mg\ell}{2}$$

$u^2 = v^2 + g\ell$

$$v = \sqrt{u^2 - g\ell}$$

$$= \sqrt{3^2 - 10 \times 0.5}$$

$= \sqrt{9-5} = 2$ m/s

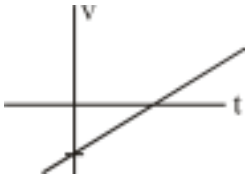
14) As $x = 8 - 4t + 6t^2$

$$\therefore v = \frac{dx}{dt} = 0 - 4 + 12t \quad \text{or} \quad v = 12t - 4$$

As t = 0, v = –4. Also $v \propto t$

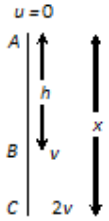
Hence, (1) represents the correct graph.

OR



$x = 8 - 4t + 6t^2$

$v = -4 + 12t$



Let at point A initial velocity of body is equal to zero

For path AB : $v^2 = 0 + 2gh$ ... (i)

For path AC : $(2v)^2 = 0 + 2gx$

$\Rightarrow 4v^2 = 2gx$ ... (ii)

15) Solving (i) and (ii) $x = 4h$

16)

Thickness = 6.5mm + 45 $\left(\frac{\frac{1}{2}mm}{50}\right)$

= 6.95 mm

17) W = Area = $\frac{1}{2}\pi r_1 r_2 = \frac{1}{2}\theta\left(\frac{x_0}{2}\right)(F^0) = \frac{\pi x_0 F_0}{4}$

18) **Explain Question:**

We have to calculate maximum compression of spring.

Concept:

Work Energy theorem.

Solution:

by WET

$\frac{1}{5} \times 50 \times (2)^2 - -0.1 \times 50 \times 10x = \frac{1}{2} \times 100x^2$

$x^2 + x - 2 = 0$

x = + 1, - 2

Compression cannot be negative so x = 1m.

Final Answer: Option (1)

19) $\frac{dm'}{dt} = \frac{4dm}{dt}$

ρA'v' = 4ρAv

2Av' = 4Av

$\boxed{v' = 2v}$

P' = ρA'v'^{3}

P' = ρ(2A) (2v)^3

P' = 16ρAv^3

$\boxed{P' = 16p}$

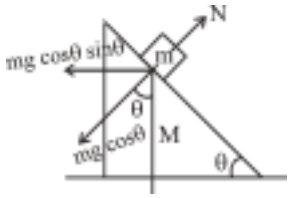
20) $u = x^2 + 2y^2 + z^2$

$$\vec{F} = \frac{-\partial u}{\partial x}\hat{i} - \frac{-\partial u}{\partial y}\hat{j} - \frac{-\partial u}{\partial z}\hat{k}$$

$$\vec{F} = -2x\hat{i} - 4y\hat{j} - 2z\hat{k}$$

$$\vec{F}\Big|_{(-1,1,-2)} = 2\hat{i} - 4\hat{j} + 4\hat{k}$$

21)

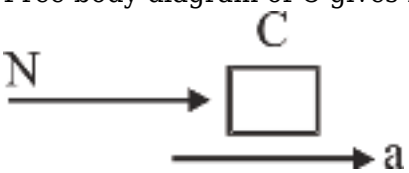


Frictional force = mg cosθ sinθ = $\frac{1}{2}mg\sin 2\theta$

22) Horizontal acceleration of the system is, $a = \frac{F}{2m + m + 2m} = \frac{F}{5m}$.

Let N be the normal reaction between B and C.

Free body diagram of C gives N = 2ma = $\frac{2}{5}F$



Now, B will not slide downward if

$\mu N \geq m_B g$

or $\mu\left(\frac{2}{5}F\right) \geqslant mg$

or $F \geq \frac{5}{2\mu}mg$

so $F_{min} = \left(\frac{5}{2\mu}\right)mg$

23) For 1 kg not slipping

$a = g \tan \theta$ ($\theta = 37°$)

$a = 10 \times 3/4$

$a = 7.5\ m/s^2$

For wedge & 1kg block

$T = (1 + 4)a$

$T = 5 \times 7.5$

$T = 37.5\ N$

for m

$mg - T = ma$

$10\ m - 37.5 = m \times 7.5$

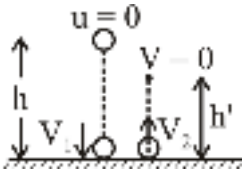
$2.5\ m = 37.5$

m = 15 kg

24) For man for block
T – mg = ma $T = 100\ g \sin 53°$
T = m(g + a) $50(g + a) = 100\ g \sin 53°$
T = 50(g + a) $g + a = 2g \times \frac{4}{5}$
$\Rightarrow a = 6\ m/s^2$

25) **Question Explanation:**
We have to calculate total distance covered by the ball before second hit.

Concept:
Inelastic collision.

Solution:


Let ball hit floor by V_1 just before first collision.
Energy conservation :
$\frac{1}{2}mV_1^2 = mgh$
$\Rightarrow V_1 = \sqrt{2gh}$ ....(1)
$e = \frac{V_2}{V_1}$
[V_2 = Velocity with which ball rises after Ist collision]
Let it attains height h' after first collision.
Energy conservation
$\frac{1}{2}mV_2^2 = mgh'$
$\Rightarrow \frac{1}{2}m(eV_1)^2 = mgh'$
$\Rightarrow \frac{1}{2}me^2(2gh) = mgh'$
$\Rightarrow h' = e^2h$ ....(2)
So, Total distance travelled before second collision = h + 2h'
$= h + 2e^2h = h(1 + 2e^2)$

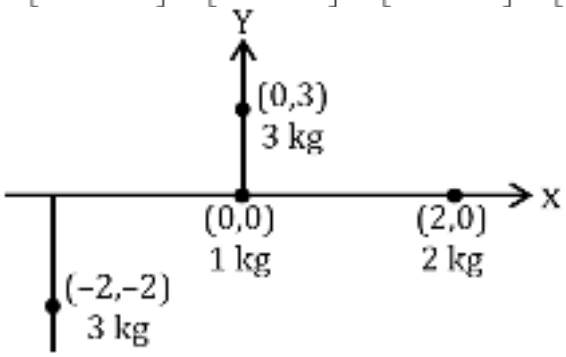
Final Answer: Option(2)

26)

Explanation :- In this question we have to find the moment of inertia of system of discrete particles.
Concept :- Moment of Inertia of discrete particle system.

Formula :- $I = I_1 + I_2 + \ldots\ldots = M_1r_1^2 + \ldots\ldots$
Calculation :- Given here
$M_1 = 1kg$, $M_2 = 2kg$, $M_3 = 3kg$, $M4 = 4kg$
given coordinate of M_1, M_2, M_3, M_4
are (0,0,0), (2,0,0), (0,3,0) (–2, –2, 0) respectively
⇒ Moment of Inertia of whole system about x-axis will be
$I = I_1 + I2 + I3 + I4 + I_1$
$\Rightarrow I = M_1r_1^2 + M2r_2^2 + M3r_3^2 + M_4r_4^2$

$$I = \left[1 \times (O)^2\right] + \left[2 \times (0)^2\right] + \left[3 \times (3)^2\right] + \left[4 \times (-2)^2\right]$$



$I = 0 + 0 + 27 + 16 = 43kg\text{-}M^2$
Conclusion : The correct answer is option 1

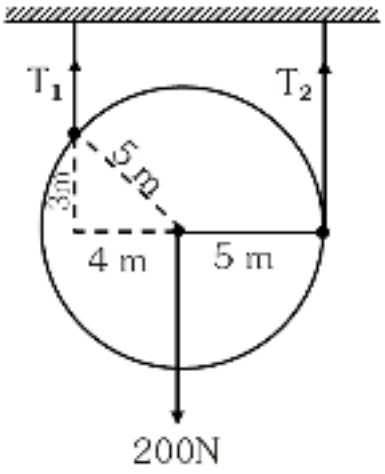
27) $L \propto \sqrt{KE}$

$$\frac{\Delta L}{L_i} = \frac{L_f - L_i}{L_i} \times 100$$

$$= \frac{\sqrt{0.49} - 1}{1} \times 100 = 30\%$$

28) $T_1 + T_2 = 200$ .....(1)
$\tau_{T_2} = 0$
$-T_1 \times 9 + 200 \times 5 = 0$
$9T_1 = 1000$

$$T_1 = \frac{1000}{9}N$$



29) By COME

PE = RE

$$mg\frac{\ell}{3} + mg\left(\frac{2\ell}{3}\right) + mg\left(\frac{3\ell}{3}\right) = \frac{1}{2} I\omega^2$$

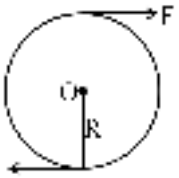
$$6\,mg\frac{\ell}{3} = \frac{1}{2}\left[\frac{m\ell^2}{9} + \frac{4m\ell^2}{9} + \frac{9m\ell^2}{9}\right].\omega^2$$

$4 \times 9\ g = 14 \square\ \omega^2$

$$\omega = 6\sqrt{\frac{g}{14\ell}}$$

$$v = \omega r = 6\sqrt{\frac{g}{14\ell}} \cdot \frac{2\ell}{3}$$

$$= \sqrt{\frac{16g\ell}{14}} = \sqrt{\frac{8g\ell}{7}}$$



30)

$\tau_{net} = 2FR$

$F_{net} = 0$

31) Kepler's second law is equivalent to conservation of angular momentum.

32)

$g' = g - \omega^2 Re \cos^2\lambda$

$$\omega = \frac{1}{\cos\lambda}\sqrt{\frac{g}{R}} \qquad \left(\cos\lambda = \frac{4}{5}\right)$$

$$\omega = \frac{5}{4}\sqrt{\frac{g}{R}}$$

33) $W = (TE)_f - (TE)_i$

$= (TE)_{7R} - (PE)_R$

$$= -\frac{GMm}{2(7R)} - \left(-\frac{GMm}{R}\right)$$

$$= \frac{13}{14}\frac{GMm}{R}$$

34)

$$\frac{\Delta g}{g} \times 100 = -\frac{d}{R} \times 100$$

$$-60 = -\frac{d}{6400} \times 100$$

$d = 3840$ km

35) $V_e = \sqrt{\frac{2GM_e}{R_e}} = 11.2$ km/s

$V_p = \sqrt{\frac{2GM_p}{R_p}} = \sqrt{\frac{2G100M_e}{4R_p}} = 5\sqrt{\frac{2GM_e}{R_e}}$

$V_p = 5 \times 11.2 = 56.0$ km/s

36) $\frac{2T}{r} = \rho gh$

Point 1 to 2



$$P_a + \frac{2T}{r} - \rho gy + \frac{2T}{r} = P_a$$

$$\rho gy = \frac{4T}{r} = 2\rho gh$$

$y = 2h$

37) **Displacement condition for floating ice:**

Weight of ice + iron = Weight of displaced water

$(m_{ice} + m_{iron})\, g = \rho_{water}\, V_{displaced}\, g$

Volume of displaced water before melting:

$$V_{displaced} = \frac{(m_{ice} + m_{iron})}{\rho_{water}}$$

Volume of water after melting: Ice melts into water of same mass: $V_{ice\ -\ melted} = \frac{m_{ice}}{\rho_{water}}$

Iron sinks and displaces:

$$V_{ice\ -\ melted} = \frac{m_{ice}}{\rho_{water}}$$

Volume of displaced water after melting

$$V'_{displaced} = \frac{m_{ice}}{\rho_{water}} + \frac{m_{ice}}{\rho_{iron}}$$

As $\rho_{iron} > \rho_{water} \Rightarrow V_{displaced} > V'_{displaced}$ Initially, ice + iron displaced more water than after melting because the iron was supported by the ice's buoyancy. After melting, the iron sinks and displaces less water than it did while floating with the ice. Since the iron displaces less water after melting, the overall **water level decreases.**

38)

$2v_1 = v_2$ .....(i)

$$\frac{1}{2}\rho(v_2^2 - v_1^2) = \rho g \Delta h$$

$$v_2^2 - v_1^2 = 2g\Delta h$$

$$4v_1^2 - v_1^2 = 2g\Delta h$$

$$v_1 = \sqrt{\frac{2}{3}g\Delta h}$$

$A_1v_1 =$

39) $v = \sqrt{2gh}$

40) $$\Delta \ell = \frac{\rho g \ell^2}{2Y} = \frac{0.75 \times 10^3 \times 10 \times 4 \times 4}{2 \times 5 \times 10^8}$$

$$\frac{12 \times 40^4}{109} = 12 \times 10^{-5} m$$

$= 1.2 \times 10^{-4}$ m

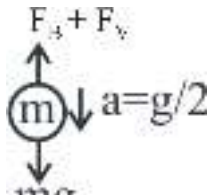
41)

$F = F_1 + F_2$

$$\frac{(2A)Y\Delta L}{L} = \frac{AY_1\Delta L}{L} + \frac{AY_2\Delta L}{L}$$

$2Y = Y_1 + Y_2$

$$Y = \frac{Y_1 + Y_2}{2}$$

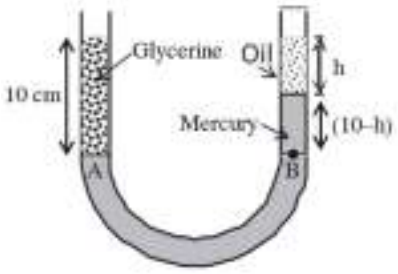


42)

$F_{net} = ma$

$mg - F_V - F_B = m\left(\frac{g}{2}\right)$

by solving $v_t = \frac{r^2 g}{9\eta}(\rho - 2\sigma)$



43)

$P_A = P_B$

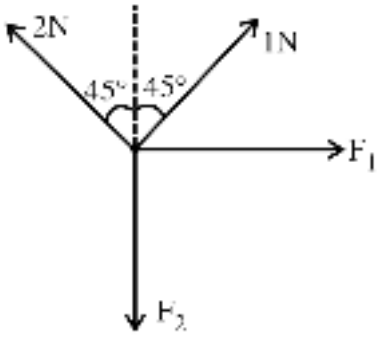
(1.3)g(10) = (0.8)gh + (13.6)g(10–h)
13 = 0.8h + 136 – 13.6h
12.8h = 123
$h = \frac{123}{12.8} = 9.6\text{cm}$

44) $a = -\alpha x^2 \Rightarrow \int_{V_0}^{0} V\,dv = -\alpha \int_{0}^{x} x^2 dx$

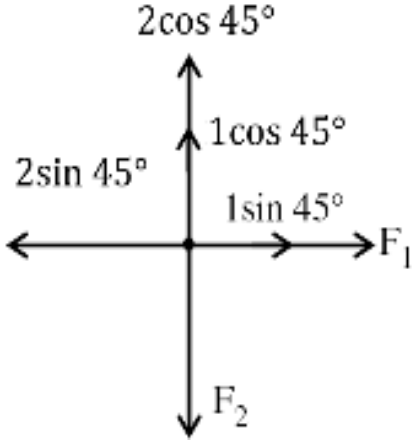
$\Rightarrow -\frac{V_0^2}{2} = -\alpha\frac{x^3}{3}$

$\Rightarrow x = \left(\frac{3V_0^2}{2\alpha}\right)^{1/3}$

45)



Resolving 1N and 2N forces into its components



In equilibrium condition

$F_1 + 1\sin 45° = 2\sin 45°$

$F_1 = \frac{2}{\sqrt{2}} - \frac{1}{\sqrt{2}} \Rightarrow F_1 = \frac{1}{\sqrt{2}}$

$F_2 = 1\cos 45° + 2\cos 45°$

$F_2 = \frac{1}{\sqrt{2}} + \frac{2}{\sqrt{2}} \Rightarrow F_2 = \frac{3}{\sqrt{2}}$

CHEMISTRY

46)

CO form in ground state

47)

KI and Na_2O have only ionic bond.

48)

$[BF_4]^-$ does not has vacant orbital.

49) O_2 → O_2^+
2 unpaired e^- 1 unpaired e^-

50)

SO_2Cl_2 $\boxed{P_\pi - P_\pi = 0}$ $\boxed{P_\pi - d_\pi = 2}$

51) $I_3^-, HgCl_2, XeF_2$ = Linear, planar, $\boxed{\mu = 0}$

52)

XeF_5 not form due to odd e^- species.

53)

XeF_6 Hybridisation = sp^3d^3
LP = 1
BP = 6

54)

N > C > P > Si (EN)

55)

$Be(OH)_2$ is amphoteric in nature

56)

$O^- + e^- \rightarrow O^{-2}$
Endo thermic due to anion to e^- repulsion

57)

$F > Cl > Cl^- > F^-$ (IE order) and size of Cl^- is larger than F^-

58)

15, 190, 210, 230 ev/atom so valence e^- is 1 and answer is Na.

59)

S to S^- formation is highly exothermic among them.

60)

Cl^- and Ca^{+2} are isoelectronic but only e^- will be same.

61)

[Kr] $5s^1$ is s-block element

62) n = 1 mole $N_2 \Rightarrow 1 \times 28 = 28$ gm
n = $\frac{7}{8}$ mole $O_2 \Rightarrow \frac{7}{8} \times 32 = 28$ gm
n = 1 mole CO = 28 gm

63) $\text{moles} = \frac{11.2}{22.4} = \frac{1}{2}$ mole
$\Rightarrow \boxed{\frac{mm}{2} = VD}$ mm = molecule mass

64) If emperical formula of 2 compounds are same then % of each element in compounds will be same

65) $C_4H_{10} + \frac{13}{2} O_2 \rightarrow 4 CO_2 + 5H_2O$
$\frac{1000}{58}$ mole
mass of $O_2 = \frac{13}{2} \times \frac{1000}{58} \times 32 = 3.58$ kg

66) $C + O_2 \rightarrow CO_2$
2 mole 3 mole
O_2 remaining = 3 – 2 = 1 mole, 32 gm

67) (A) 2 N_A molecule
(B) 1 N_A molecule
(C) 1 N_A molecule
(D) 0.5 N_A molecule

68) $N_2 + 3H_2 \rightarrow 2NH_3$
1 3
$\frac{v_f}{v_i} = \frac{2}{4} = \frac{1}{2}$
$\Rightarrow \frac{v_i}{v_f} = \frac{2}{1}$

69) more the energy released strong will be acid

70) $\Delta H = 42 = (\Delta H_f)_P - (\Delta H_f)_R$
$42 = -94 - 152 - (\Delta H_f CaCO_3)$
$\Rightarrow \Delta H_f\ CaCO_3 = -288$ kJ

71) $H_{2(g)} + \frac{1}{2} O_{2(g)} \xrightarrow{-68.32} H_2O_{(\square)} \xrightarrow{10.52} H_2O_{(g)}$
$(\Delta H_f)\ H_2O_{(g)} = -68.32 + 10.52$
$= -57.80$

72) $\Delta H_{reaction} = (\Delta H_f)_P - (\Delta H_f)_R$
$= -24.8 - [-17.9 + 12.5]$
$= -19.4$ kcal

73) $Fe + S \rightarrow FeS$
2.1 gm
$\frac{2.1}{56}$ mole
$\frac{2.1}{56}$ mole Fe $\Rightarrow -3.75$ kJ
One mole Fe $\Rightarrow \frac{-3.75 \times 56}{2.1} = -100$ kJ

74) $A_{(g)} \rightleftharpoons B_{(g)} + C_{(g)}$

	1	0	0
	1–x	x	x
mole fraction	$\frac{1-x}{1+x}$	$\frac{x}{1+x}$	$\frac{x}{1+x}$

$PP_A = \frac{(1-x)}{1+x} P = \frac{P}{2}$

$\Rightarrow x = \frac{1}{3}$

$K_P = \frac{\frac{P}{4} \cdot \frac{P}{4}}{\frac{P}{2}} = \frac{P}{8}$

75) $CaCO_3$ ⇌ CaO + CO_2
(s) (s) (g)
$\frac{1}{5}$ mole 0 0
$\frac{1}{5} - \frac{65}{100} \times \frac{1}{5}$ $\frac{65}{100} \times \frac{1}{5}$ $\frac{65}{100} \times \frac{1}{5}$
PV = nRT

$P = \frac{65}{100} \times \frac{1}{5} \times \frac{1}{12} \times \frac{1092}{10}$

P = 1.183 atm = $K_P = P_{CO_2}$

76) $\Delta E = 13.6z^2 = \left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$

$\Delta E = 13.6 \times 4 \times \left[\frac{1}{1^2} - \frac{1}{3^2}\right] = 48.36$ eV

77) Isotopes ⇒ protons same = Z
Isodiaphers ⇒ A – 2Z = same
Isotones ⇒ A – Z = same
Isoelectronic ⇒ e^- = same

78) $\Delta E = 13.6z^2 = \left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$

$\frac{\Delta E_1}{\Delta E_2} = \frac{13.6 \times 4\left[\frac{1}{2^2} - \frac{1}{3^2}\right]}{13.6 \times 9\left[\frac{1}{1^2} - \frac{1}{2^2}\right]}$

$\frac{\Delta E_1}{\Delta E_2} = \frac{20}{243}$

79) $\Delta E = 13.6z^2\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$

$114.75 = 13.6z^2\left[\frac{1}{1^2} - \frac{1}{(4)^2}\right]$

$Z^2 = 9$
Z = 3

80) [HCOOH] = [HCOONa]

$pH = pK_a + \log \frac{[Salt]}{[Acid]}$

$pH = pK_a$

$[H^+] = K_a = 2 \times 10^{-4}$

81) $[H^+] = \frac{0.02 \times 25 + 0.01 + 25}{50} = \frac{0.75}{50}$

$= 0.015$ M

$[H^+] \uparrow$ pH↓

82) $FCH_2COOH \rightleftharpoons FCH_2COO^- + H^+$

$C(1-\alpha) \quad C\alpha \quad C\alpha$

$K_a = \frac{C\alpha . C\alpha}{C(1-\alpha)}$

$2.7 \times 10^{-3} = 1.5 \times 10^{-3} \times \frac{\alpha}{1-\alpha}$

$\frac{\alpha}{1-\alpha} = \frac{9}{5}$

$\alpha = \frac{9}{14}$

$C\alpha = 1.5 \times 10^{-3} = [H^+]$

$C = \frac{1.5 \times 10^{-3}}{\alpha}$

$C = 2.33 \times 10^{-3}$ M

83) $[H^+] = 40 \times 0.1 = 4$

$[OH^-] = 10 \times 0.45 = 4.5$

$[OH^-] = \frac{4.5-4}{40+10} = \frac{0.5}{50} = \frac{1}{100} = 10^{-2}$

⇒ pOH = 2

pH = 12

84) Relative strength

$\sqrt{\frac{K_{a_1} C_1}{K_{a_2} C_2}} = \sqrt{\frac{K_{a_1}}{K_{a_2}}} = \sqrt{\frac{20}{1}} = \frac{4.47}{1}$

85) $K_{sp} = [Ag^+]^2 [SO_4^{2-}]$

$6.4 \times 10^{-5} = (0.1)^2 (SO_4^{2-})$

$[SO_4^{2-}] = 6.4 \times 10^{-3}$ M

86) NaOH ⇒ SB, pH is defined from strong Base in a mixture of SB & WB

$[OH^-] = 0.1$, pOH = 1, pH = 13

87) Oxidation state of F is always –1

88) NH_4^+ & NO_3^- ⇒ NH_4^+ ⇒ N = –3, NO_3^- ⇒ N = +5

89) gm eq. HNO_3 = gm eq. I_2

$$\frac{m}{63} \times 1 = \frac{5}{254} \times 10$$

$$m = \frac{63 \times 50}{254}$$

m = 12.4 gm

90) gm eq. $H_2C_2O_4$ = gm eq $KMnO_4$

$$0.1 \times V = \frac{20}{158} \times 5$$

$$V = \frac{1000}{158} = 6.3 \text{ L}$$

BIOLOGY

91)

NCERT XI Pg : 5 (Old NCERT)

92) NCERT Pg # 6

93) NCERT-XI Pg # 24

94) NCERT-XI, Pg # 4

95)

NCERT XI, Page No. # 6,7
A,B & D are correct

96)

NCERT Pg. # 11

97) NCERT-XI, Pg. # 91

98) XI New NCERT Page. No. 15

99)

NCERT XI Pg # 17

100)

NCERT Pg.# 20, 21

101) NCERT Pg. # 17, 18

102) NCERT Pg. # 15, 16

103) NCERT Page No.# 21

104) NCERT Pg.13, 14, 15

105) NCERT Pg. 14, 15, 16

106)

NCERT (XI) Pg # 24

107) NCERT-XII Page No. 26

108) NCERT XI Page No. # 30

109)

NCERT Pg. # 32

110) NCERT-XI Page No. 32-33

111)

NCERT Pg. # 13, 25

112)

NCERT-XI, Pg. # 28, 32, 33

113) NCERT-XI, Pg. # 30

114)

A and D are correct

115)

NCERT XI Pg. # 93, 94
B and C are incorrect

116)

NCERT-XI, Pg. # 97

117)

NCERT-XI, Pg. # 97, 98

118)

NCERT-XI, Pg.# 101

119)

NCERT-XI, Pg. # 101

120) **Explanation** : The question asks which enzyme in peroxisomes converts hydrogen peroxide (H_2O_2) into water (H_2O) and oxygen (O_2)
Concept : Micobodies - peroxisomes
Solution :

A. Peroxisomes contain oxidative enzymes.

B. Hydrogen peroxide (H_2O_2) is a by-product of oxidative reactions.

C. Catalase is the enzyme that breaks down H_2O_2 into harmless water and oxygen.

D. $2H_2O_2 \xrightarrow{\text{catalase}} 2H_2O + O_2$

Final Answer : Catalase

121) NCERT-XI, Pg. # 96

122)

NCERT-XI, Pg. # 96

123)

NCERT-XII Page No. # 121-122

124) NCERT XII Page No. # 122, 123 & 124

125) NCERT XI Page No. # 126

126) NCERT, Pg. # 121

127) Module-6, Pg # 210

128) NCERT-XI, Pg # 122

129) NCERT Pg. No. # 126

130) NCERT-XI, Pg. # 106

131) NCERT Pg. # 109

132)

NCERT Pg. # 110

133)

NCERT Pg. # 106

134) NCERT Pg. # 117

135)

NCERT Pg. # 113

136)

NCERT XI, Pg # 54

137)

NCERT Pg. # 41

138)

NCERT Pg. # 47, 48

139)

NCERT Pg. # 49

140) NCERT, Pg # 43-46

141)

NCERT Pg. # 39

142) NCERT Pg. # 49-52 (E/H)

143)

NCERT Pg. # 47

144)

NCERT Pg. # 47

145) NCERT Pg.#45.

146) NCERT Pg.#112 (Module).

147) NCERT Pg#111,112,114

148)

NCERT (Old) Pg. # 112

149) NCERT (XII) ; Page No. # 118, 121

150) NCERT Pg.# 82

151) NCERT Pg.# 82

152) NCERT-XI, Pg. # 82

153) NCERT Pg. # 116-120

154)

NCERT (Old) Pg. # 102

155)

NCERT (Old) Pg. # 102

156)

OLD NCERT-XI, Pg. # 101

157) Old NCERT, Pg. # 103

158) NCERT, Pg. # 199, 200

159)

NCERT Pg. # 194

160) NCERT, Pg. # 199-200

161) NCERT, Pg. # 203

162) NCERT Pg.#201

163) NCERT Pg # 194

164) NCERT_Pg. No. # 198

165) NCERT Pg # 197

166) NCERT, Pg. # 187, 188

167) NCERT, Pg. # 183

168) NCERT, Pg. # 184, 187, 189, 190

169) NCERT, Pg. # 183, 185

170) NCERT Pg.#184

171) NCERT Pg. No. 186

172) NCERT Page No. 189

173) NCERT, Pg. # 205

174) NCERT, Pg. 208

175) NCERT, Pg. # 213

176) NCERT, Pg. # 207

177) NCERT, Pg. # 210-212

178) NCERT Pg.#209,210

179) NCERT Pg. # 209

180) NCERT Pg. # 811